

IIC2026

Visualización de Información

Alessio Bellino

Preguntas e ingredientes para una buena visualización

~~Preguntas e ingredientes para una buena visualización~~

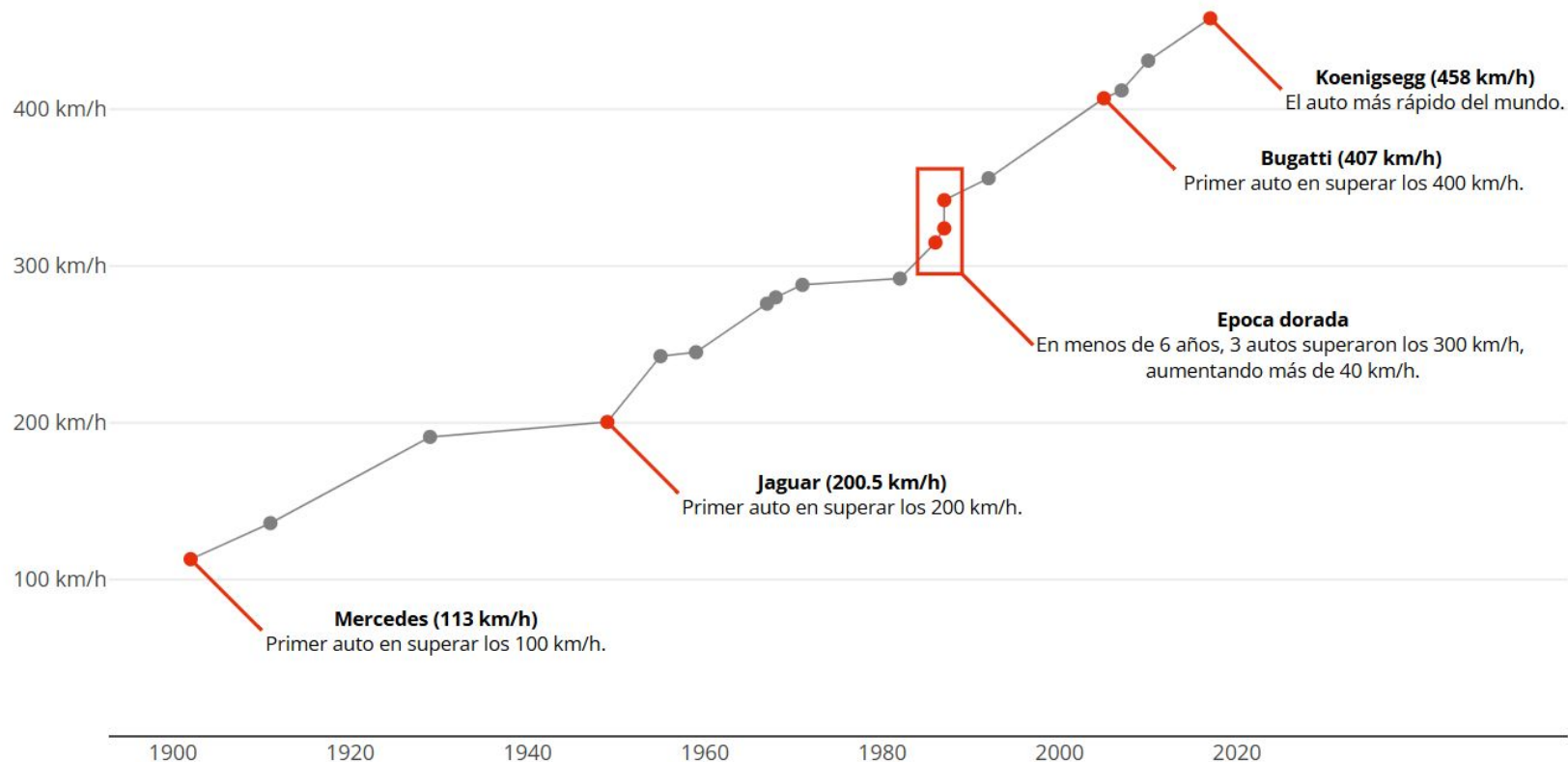
Antes de comenzar

Excelencia gráfica

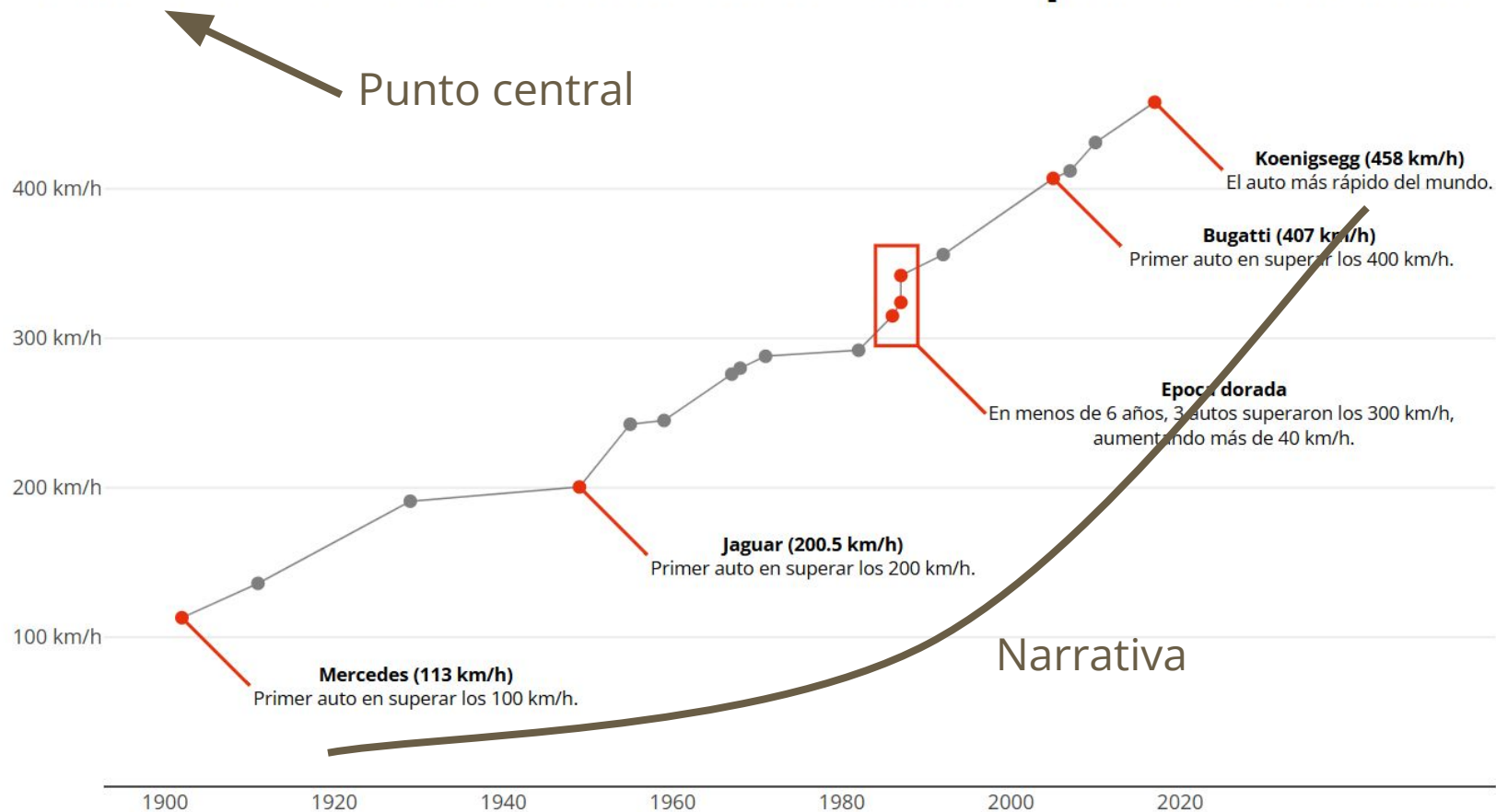
Sustancia, estadísticas y diseño

- **Definición del Punto Central:** La visualización debe tener un punto claro y bien definido que guíe la narrativa. Identificar este punto central es crucial para mantener la coherencia y el enfoque en la historia que se quiere contar.
- **Construcción de la Narrativa:** Desarrolla una historia que el espectador pueda seguir fácilmente. La historia detrás de los datos – estadísticas – debe dar sentido a la visualización, transformando gráficos en una narrativa envolvente que facilite la comprensión.
- **Impacto y Relevancia:** Asegúrate de que el diseño de la narrativa sea relevante para tu audiencia y que el mensaje principal sea claro. Una historia bien contada hace que los datos sean accesibles y memorables.

Avance de los autos comerciales más rápidos del mundo



Avance de los autos comerciales más rápidos del mundo



Claridad, precisión y eficiencia

- **Claridad:** Comunicar ideas complejas de forma fácil de entender. La visualización debe ser intuitiva y evitar ambigüedades, proporcionando una experiencia satisfactoria al usuario.
- **Precisión:** Asegurar que la información se presenta de manera exacta. Cada elemento gráfico debe tener un propósito claro y representar los datos correctamente para reforzar el mensaje.
- **Eficiencia:** Maximizar la cantidad de información transmitida con el menor desperdicio de píxeles y espacio. Basado en el "ink ratio" de Tufte, utiliza la tinta de manera efectiva para mostrar datos significativos, optimizando la comunicación.

Sobre la Eficiencia

Mayor número
de ideas posible

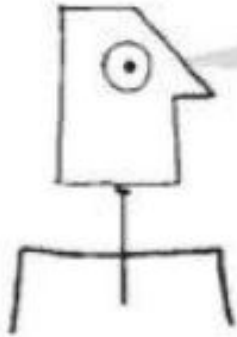


ideas

En el menor
tiempo posible



time



Usando meno
tinta posible

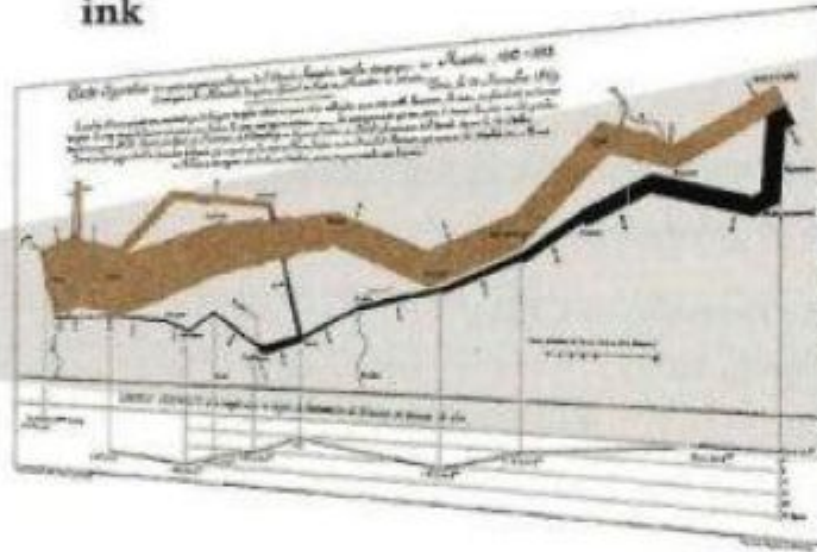


ink

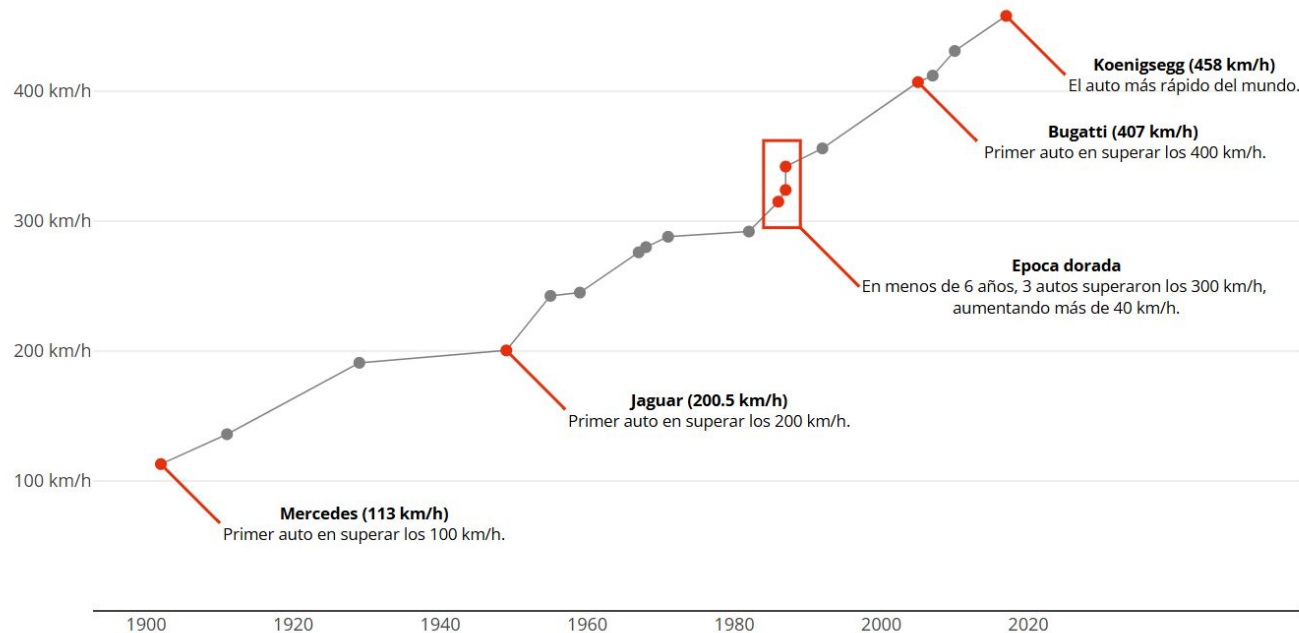
En el menor
espacio
posible



space



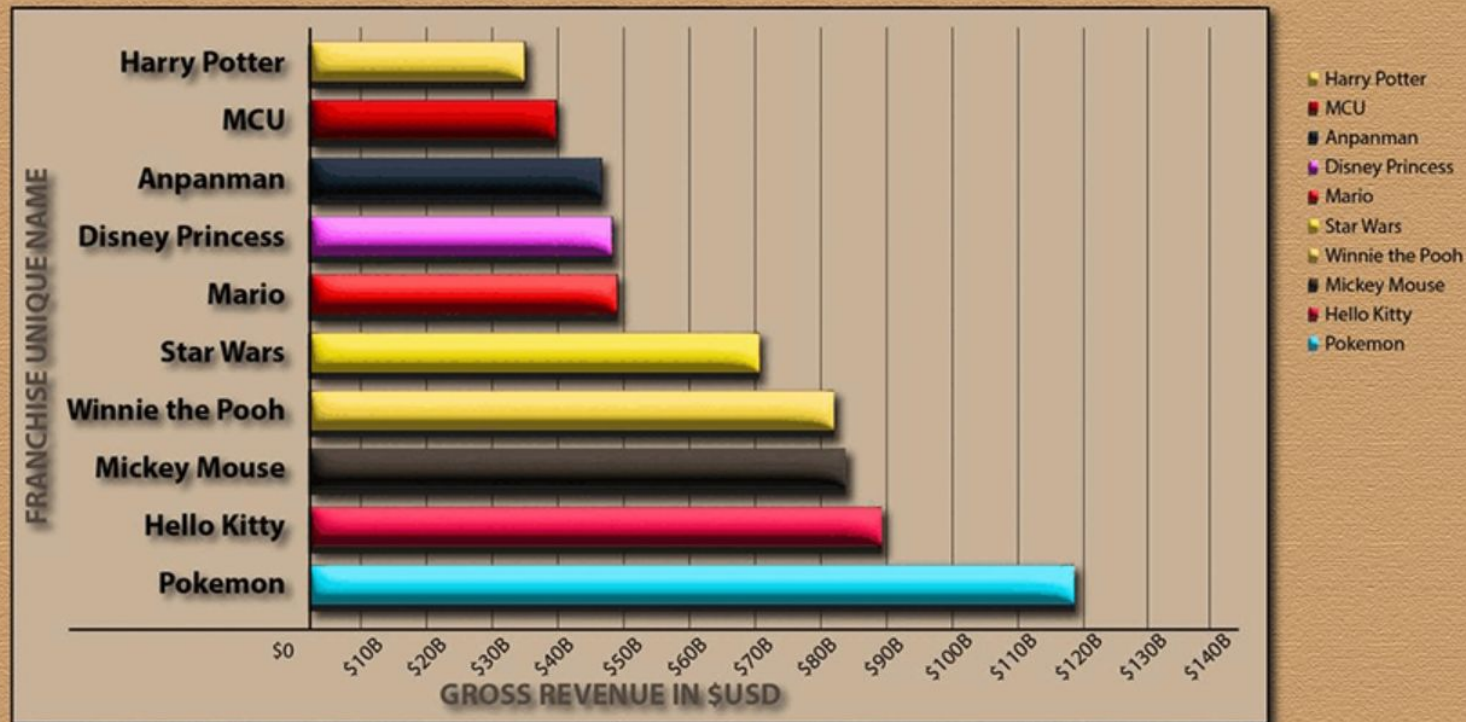
Avance de los autos comerciales más rápidos del mundo



No se gastan píxeles innecesariamente.
Cada detalle refuerza la comprensión.

Japanese Franchises Top the highest grossing media franchises of all time

REMOVE BACKGROUND



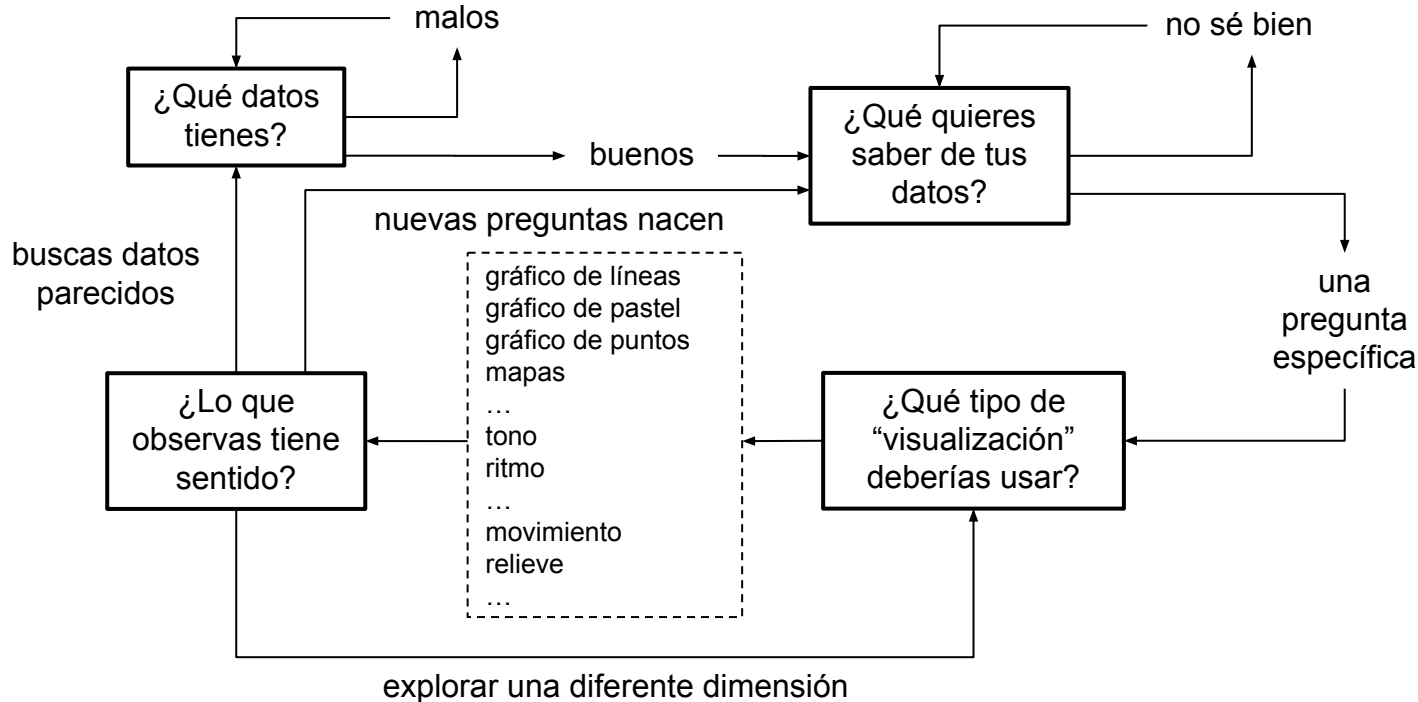
From the course PRACTICAL DATA VISUALIZATION at [DeliveringDataAnalytics.com](https://deliveringdataanalytics.com)



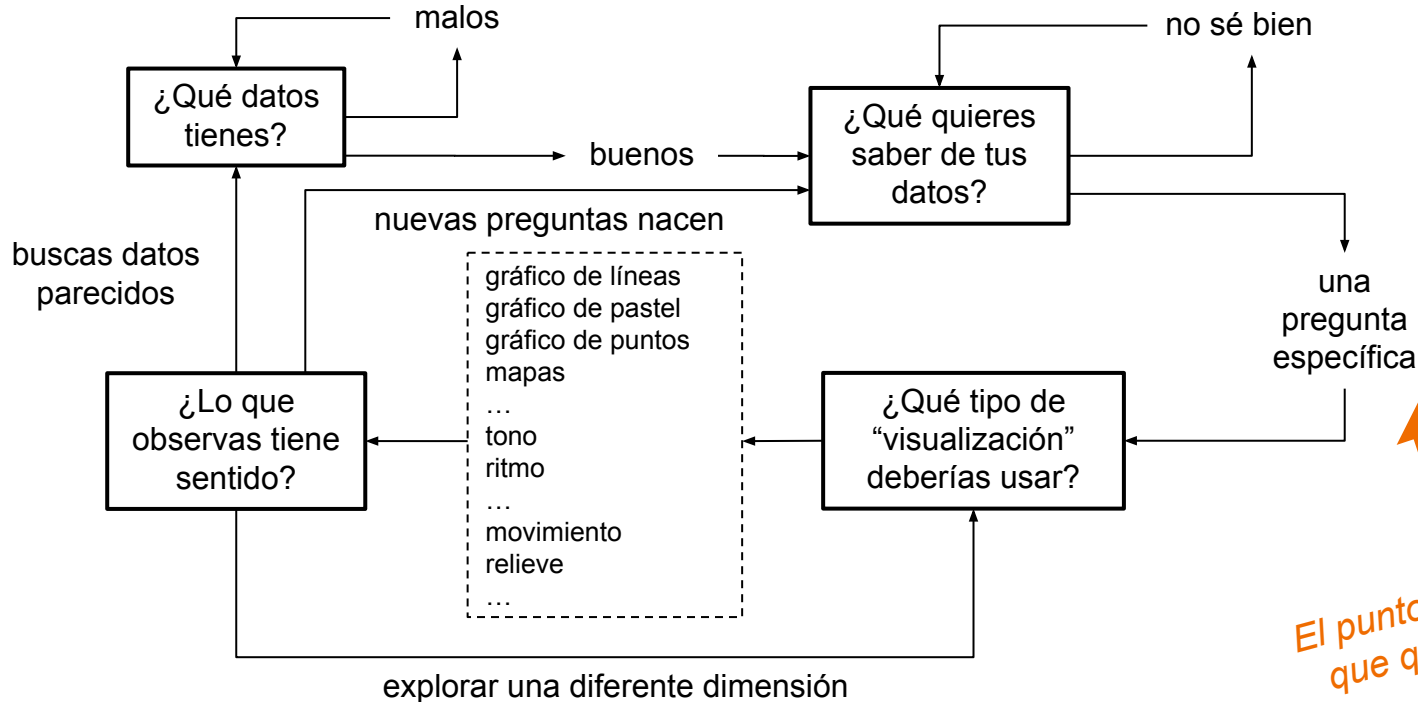
Preguntas e ingredientes

¡Comencemos!

Las preguntas adecuadas

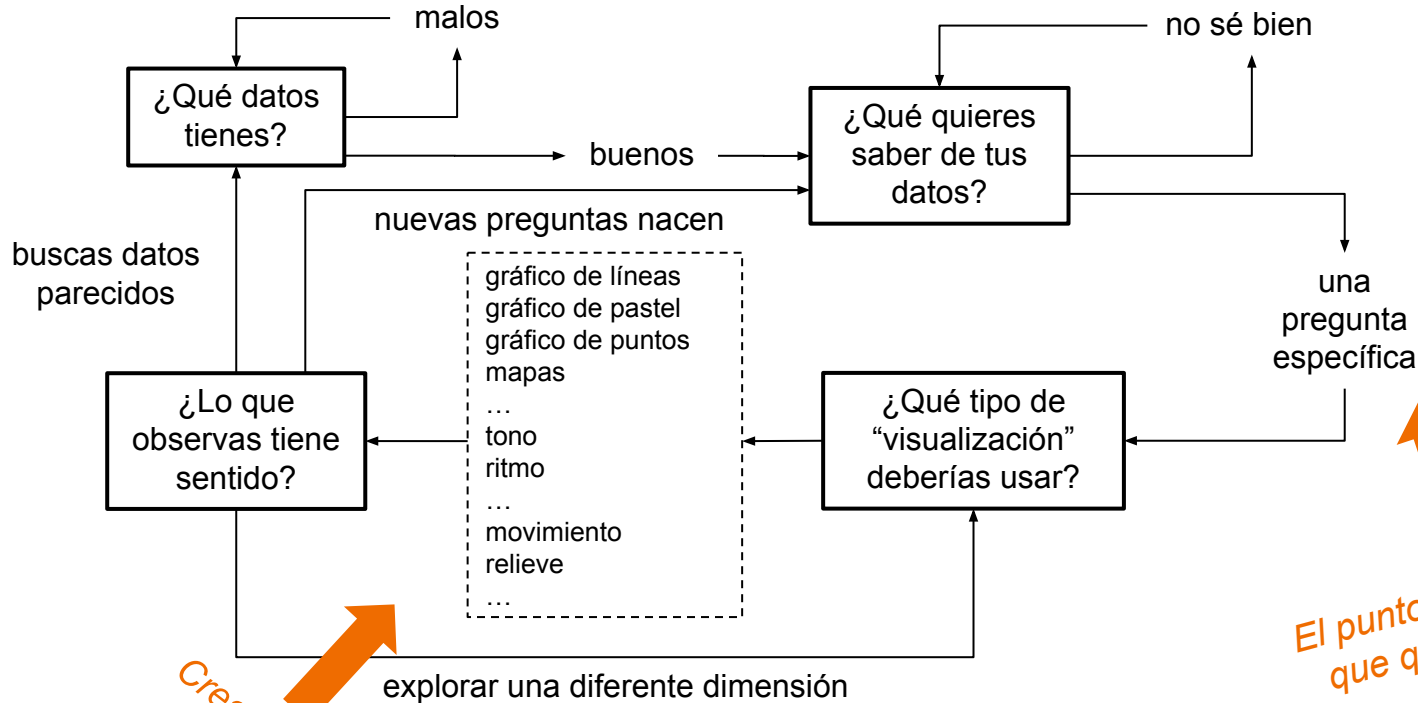


Las preguntas adecuadas

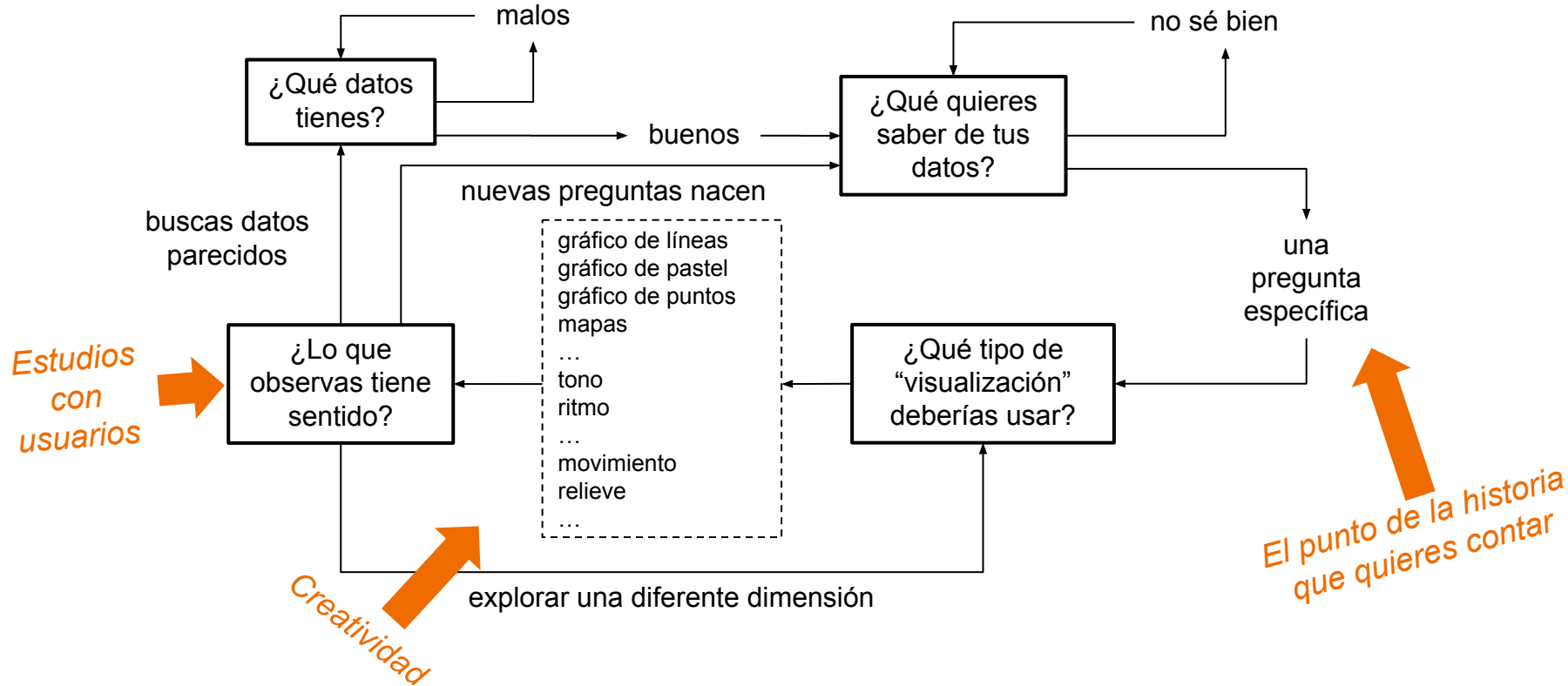


*El punto de la historia
que quieres contar*

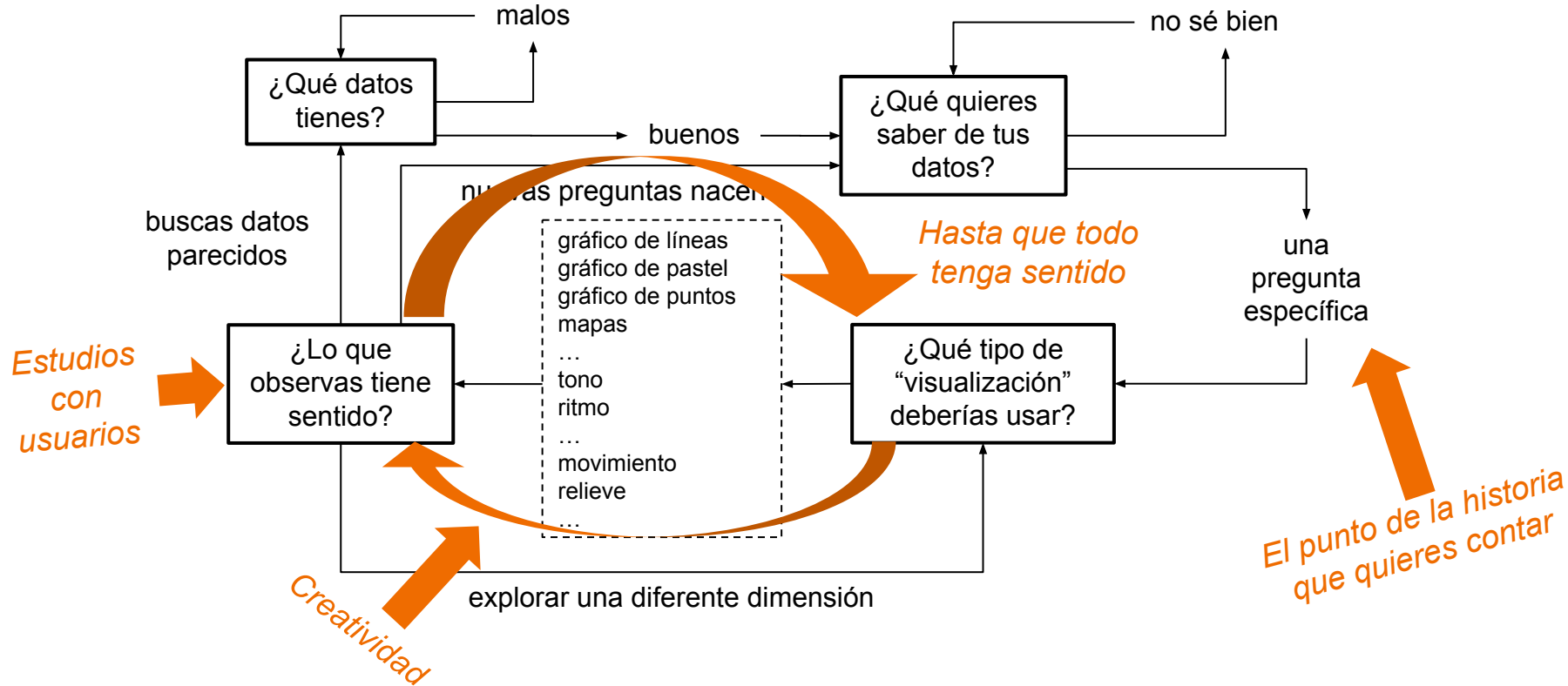
Las preguntas adecuadas



Las preguntas adecuadas



Las preguntas adecuadas



¿Qué datos existen?

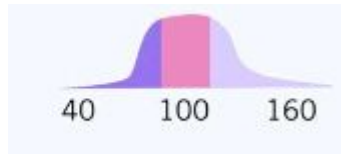
Datos numéricos

Escala de Intervalo: *sin 0 absoluto*

Temperatura °C



IQ score



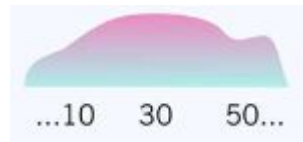
Escala de Razón: *con 0 absoluto*

Peso en KG



(ejemplo de dato continuo)

Alumnos en una clase



(ejemplo de dato discreto)

Datos categóricos

Se usa la frecuencia para su visualización

Escala Ordinal

Nivel de educación



Experiencia laboral



Escala Nominal

Smartphone vendidos



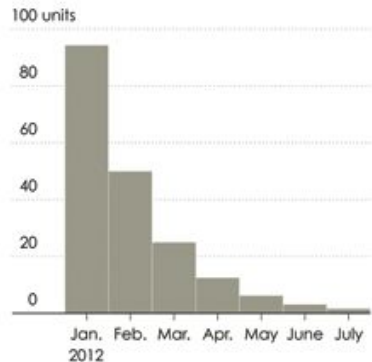
Medio de transporte usados



Los ingredientes adecuados

Title of this Graph

A description of the data or something worth highlighting to set the stage.



Source: Somewhere reputable

Title of this Graph

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Visual Cues

Visualization involves encoding data with shapes, colors, and sizes. Which cues you choose depends on your data and your goals.

Coordinate System

You map data differently with a scatterplot than you do with a pie chart. It's x- and y-coordinates in one and angles with the other; it's cartesian versus polar.

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Scale

Increments that make sense can increase readability, as well as shift focus.

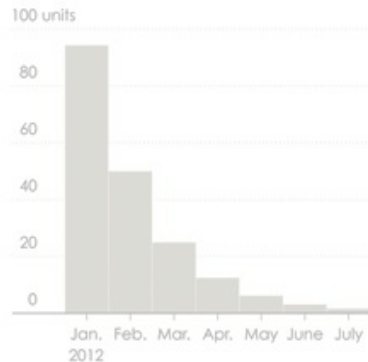
Context

If your audience is unfamiliar with the data, it's your job to clarify what values represent and explain how people should read your visualization.

Los ingredientes adecuados

Title of this Graph

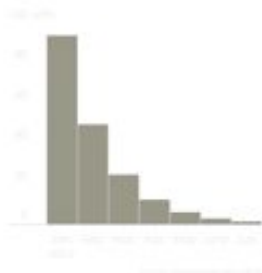
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Source: Somewhere reputable

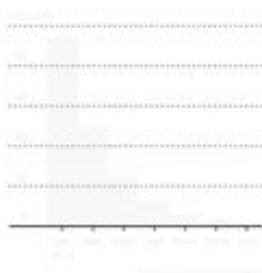
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Como mapear números visualmente

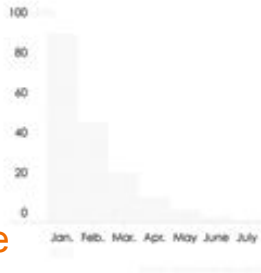
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Donde, en el espacio

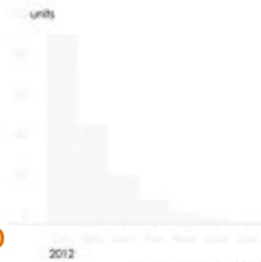
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Referencias para interpretar correctamente

Context

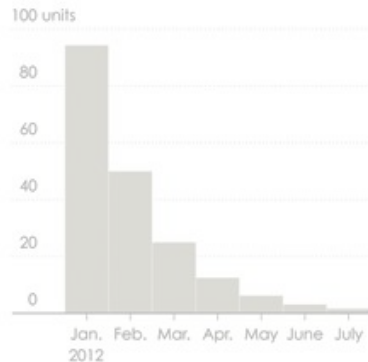
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De que estoy hablando

Los ingredientes adecuados

Title of this Graph

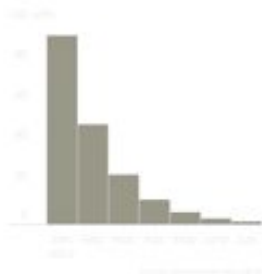
A description of the data or something worth highlighting to set the stage.



Source: Somewhere reputable

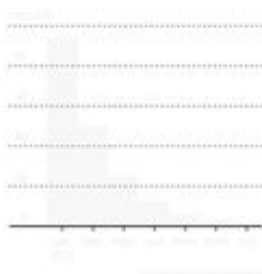
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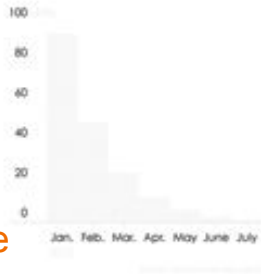
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Donde, en el espacio

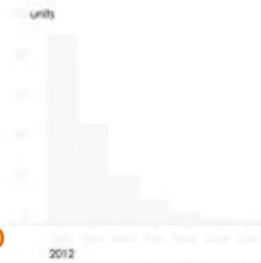
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Context

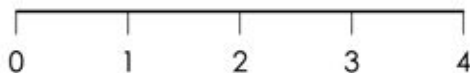
If your audience is unfamiliar with the data, it's your job to clarify what values represent and explain how people should read your visualization.

De que estoy hablando

Escalas

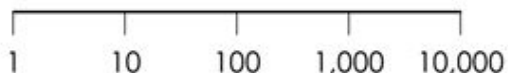
Linear

Values are evenly spaced



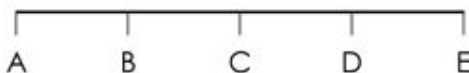
Logarithmic

Focus on percent change



Categorical

Discrete placement in bins



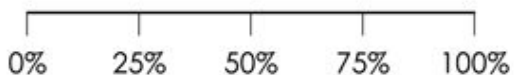
Ordinal

Categories where order matters



Percent

Representing parts of a whole



Time

Units of months, days, or hours

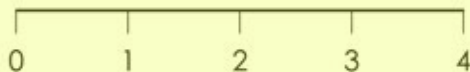


Escalas

Datos numéricos

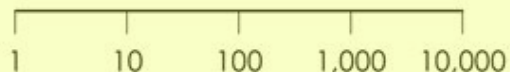
Linear

Values are evenly spaced



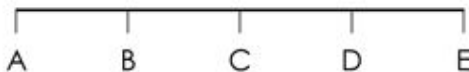
Logarithmic

Focus on percent change



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Discrete placement in bins



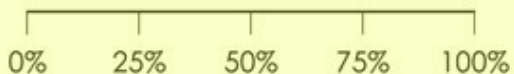
Ordinal

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Percent

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Time

Units of months, days, or hours

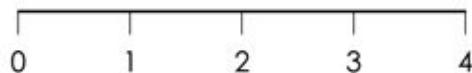


Escalas

Datos categóricos

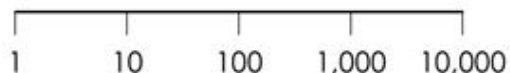
Linear

Values are evenly spaced



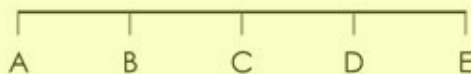
Logarithmic

Focus on percent change



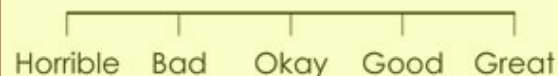
Categorical

Discrete placement in bins



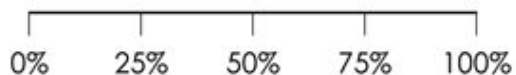
Ordinal

Categories where order matters



Percent

Representing parts of a whole



Time

Units of months, days, or hours

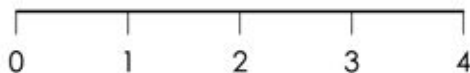


Escalas

Ambos
dependiendo del
contexto

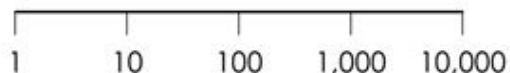
Linear

Values are evenly spaced



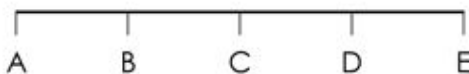
Logarithmic

Focus on percent change



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Discrete placement in bins



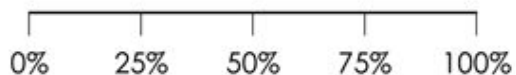
Ordinal

Categories where order matters



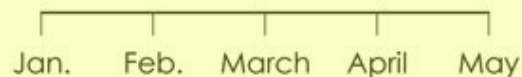
Percent

Representing parts of a whole



Time

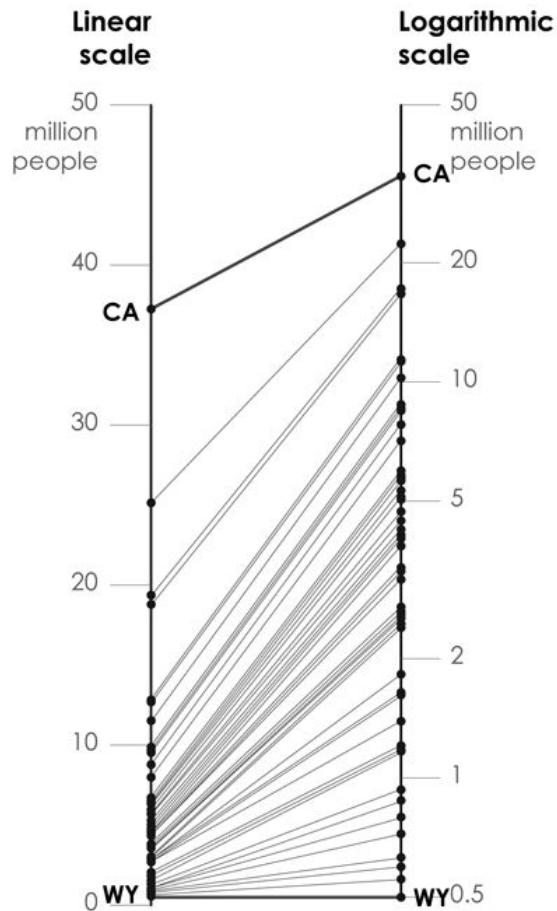
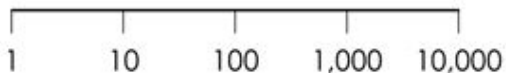
Units of months, days, or hours



Escalas logarítmicas

Logarithmic

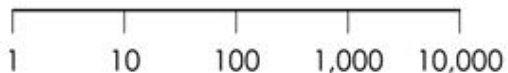
Focus on percent change



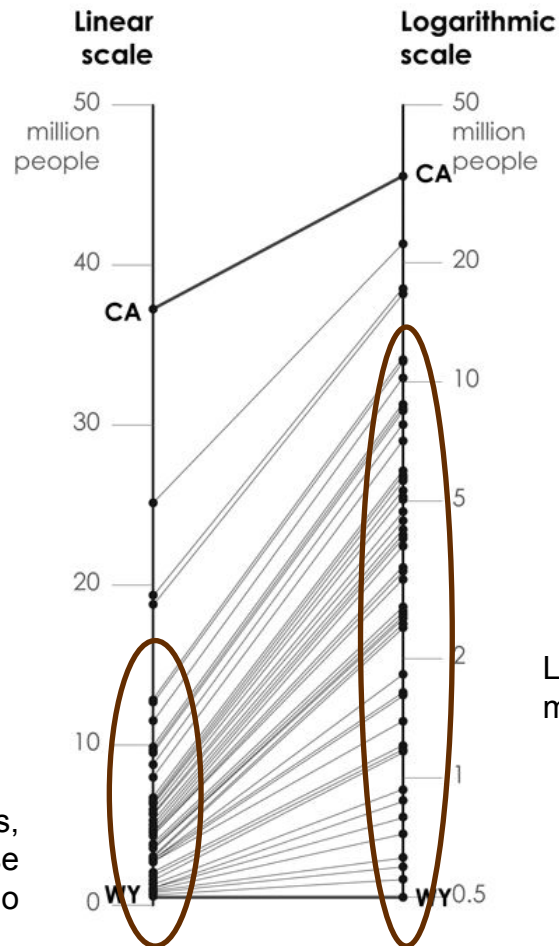
Escalas logarítmicas

Logarithmic

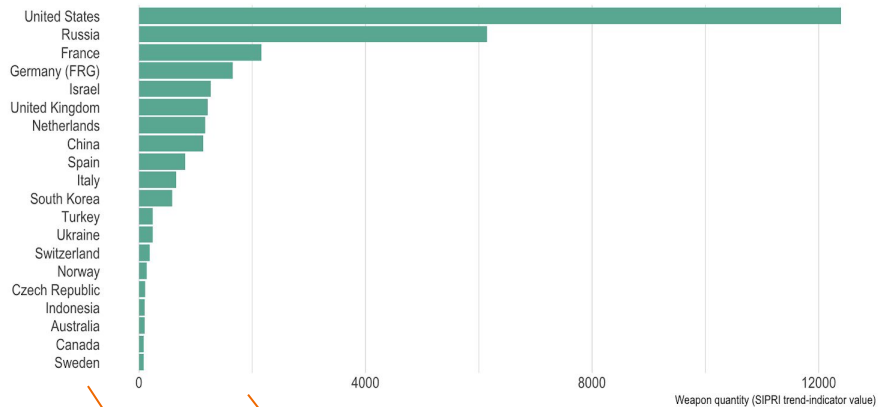
Focus on percent change



Hay muchos valores pequeños,
muy concentrados y no se
entiende mucho



Ejemplo de escalas



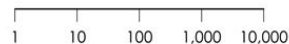
Linear

Values are evenly spaced



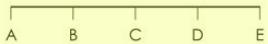
Logarithmic

Focus on percent change



Categorical

Discrete placement in bins



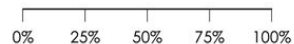
Ordinal

Categories where order matters



Percent

Representing parts of a whole



Time

Units of months, days, or hours



Ejemplo de escalas

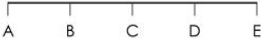
Linear

Values are evenly spaced



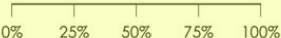
Categorical

Discrete placement in bins



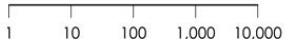
Percent

Representing parts of a whole



Logarithmic

Focus on percent change



Ordinal

Categories where order matters

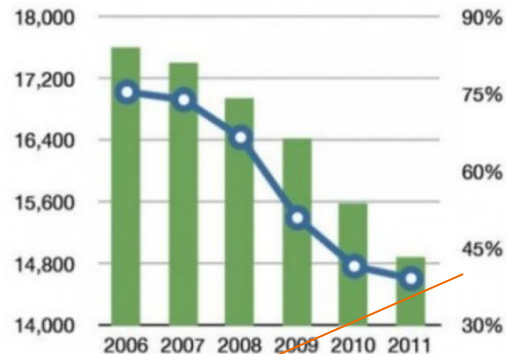


Time

Units of months, days, or hours



Internet Explorer vs Murder Rate



● Murders in US ■ Internet Explorer Market Share

Ejemplo de escalas

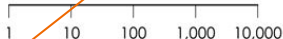
Linear

Values are evenly spaced



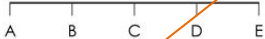
Logarithmic

Focus on percent change



Categorical

Discrete placement in bins



Ordinal

Categories where order matters



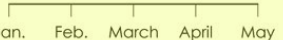
Percent

Representing parts of a whole *



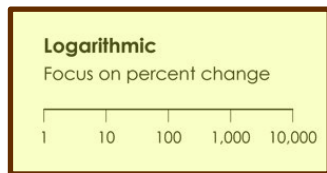
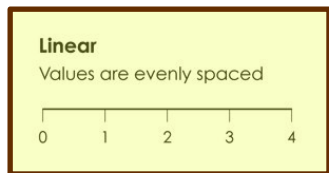
Time

Units of months, days, or hours



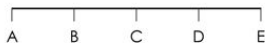
*En este caso, no se representa una parte del todo, sino un cambio porcentual.

Ejemplo de escalas



Categorical

Discrete placement in bins



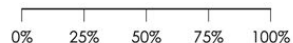
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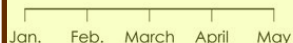
Percent

Representing parts of a whole

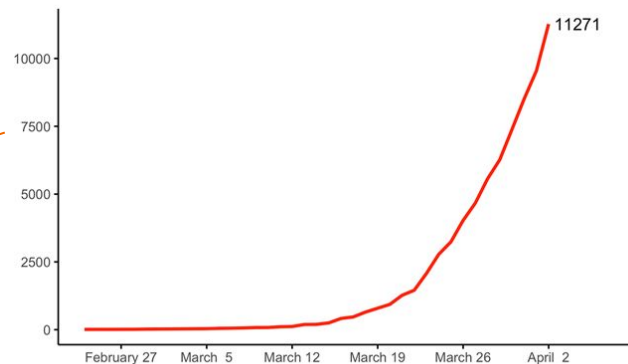


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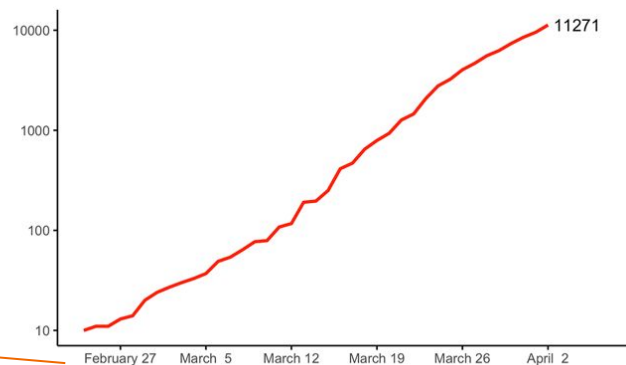


Total number of COVID-19 cases in Canada



Source: Johns Hopkins University (CSSE)

Total number of COVID-19 cases in Canada

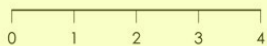


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Ejemplo de escalas

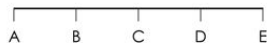
Linear

Values are evenly spaced



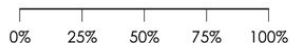
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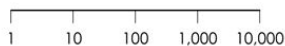
Percent

Representing parts of a whole



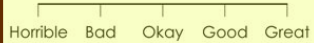
Logarithmic

Focus on percent change



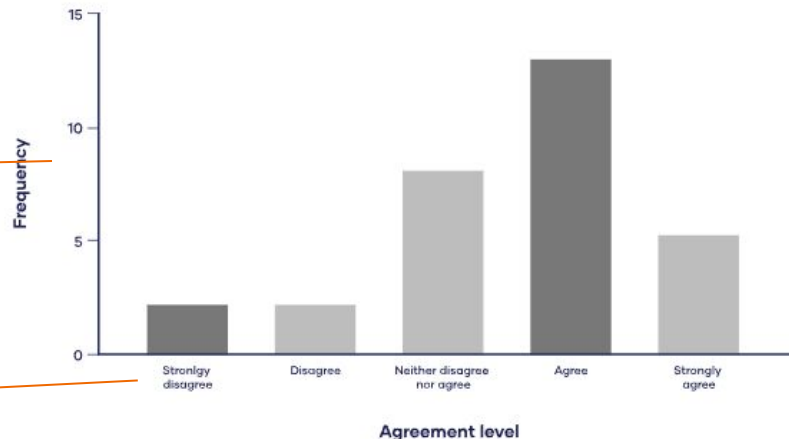
Ordinal

Categories where order matters



Time

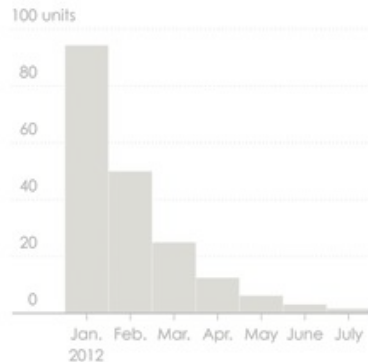
Units of months, days, or hours



Los ingredientes adecuados

Title of this Graph

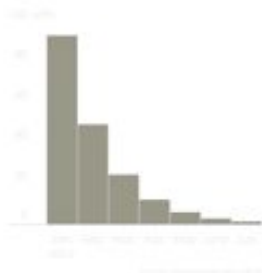
A description of the data or something worth highlighting to set the stage.



Source: Somewhere reputable

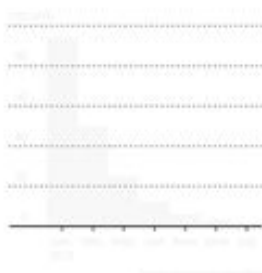
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Visual Cues

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Como mapear números visualmente

Coordinate System

You map data differently with a scatterplot than you do with a pie chart. It's x- and y-coordinates in one and angles with the other; it's cartesian versus polar.

Donde, en el espacio

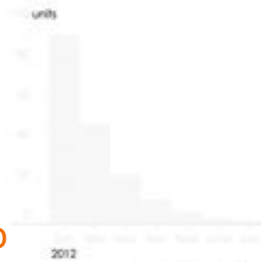
Title of this Graph

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Scale

Increments that make sense can increase readability, as well as shift focus.

Referencias para interpretar correctamente

Context

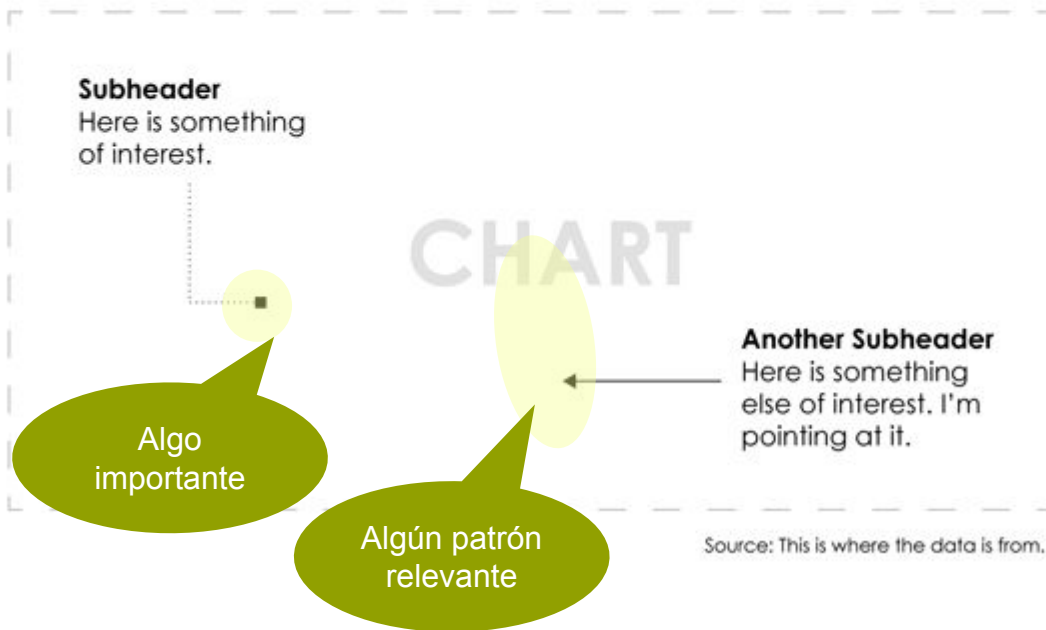
If your audience is unfamiliar with the data, it's your job to clarify what values represent and explain how people should read your visualization.

De que estoy hablando

Contexto

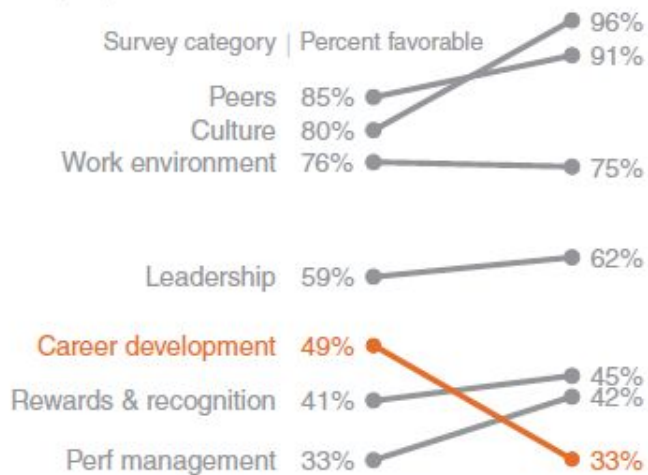
Header title that describes findings

Lead-in text is your chance to provide more details on what the data is about, where it's from, and what the audience should see or look at.



Contexto

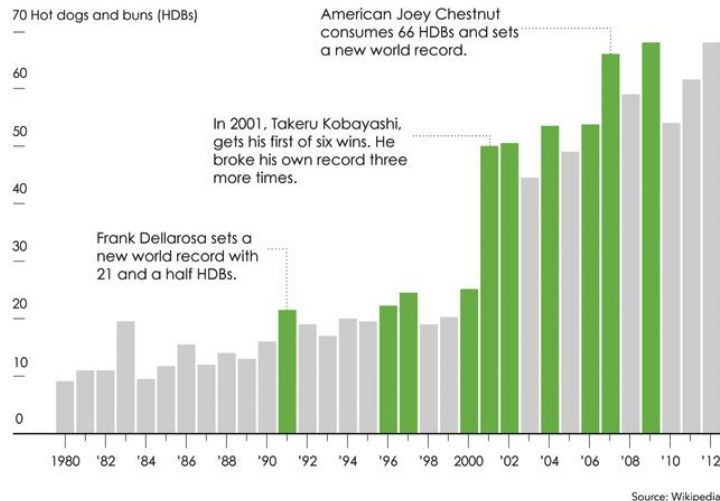
Employee feedback over time



Mostrar los datos
Evidenciar algo

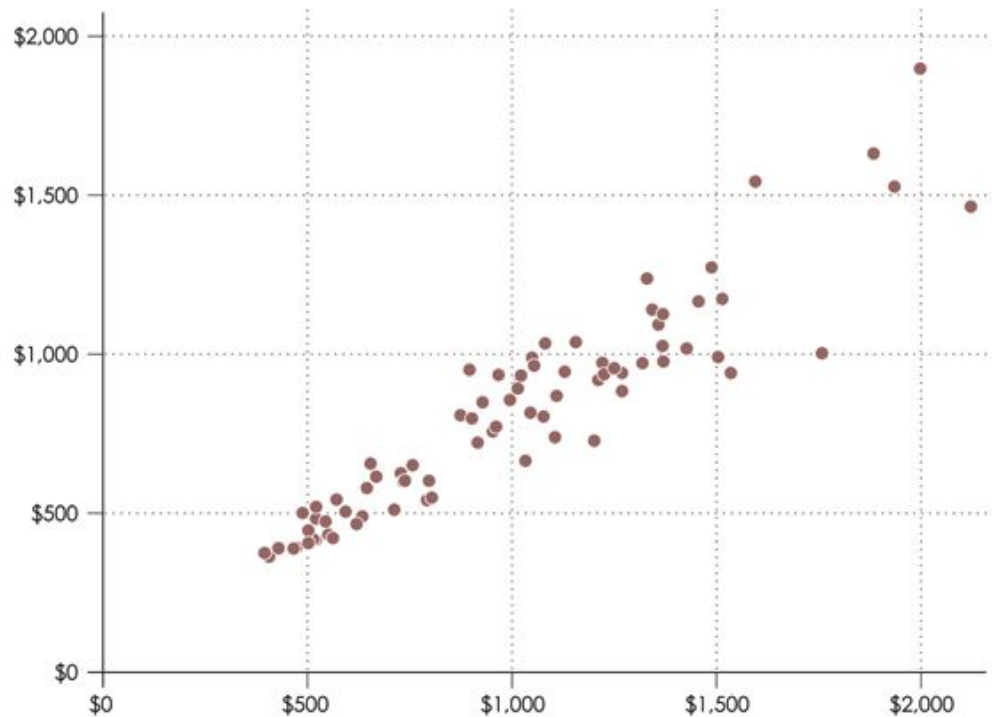
Integrar texto y gráficos

Breaking hot dog eating records



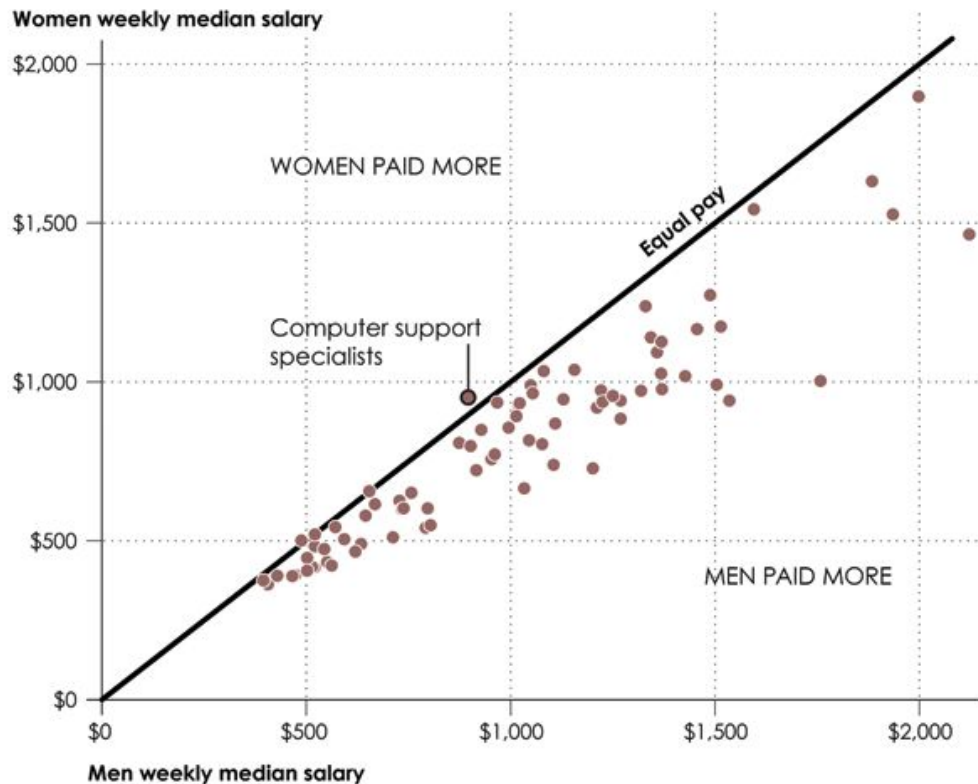
Source: Wikipedia

Contexto



Contexto

Gender pay gap in 2011

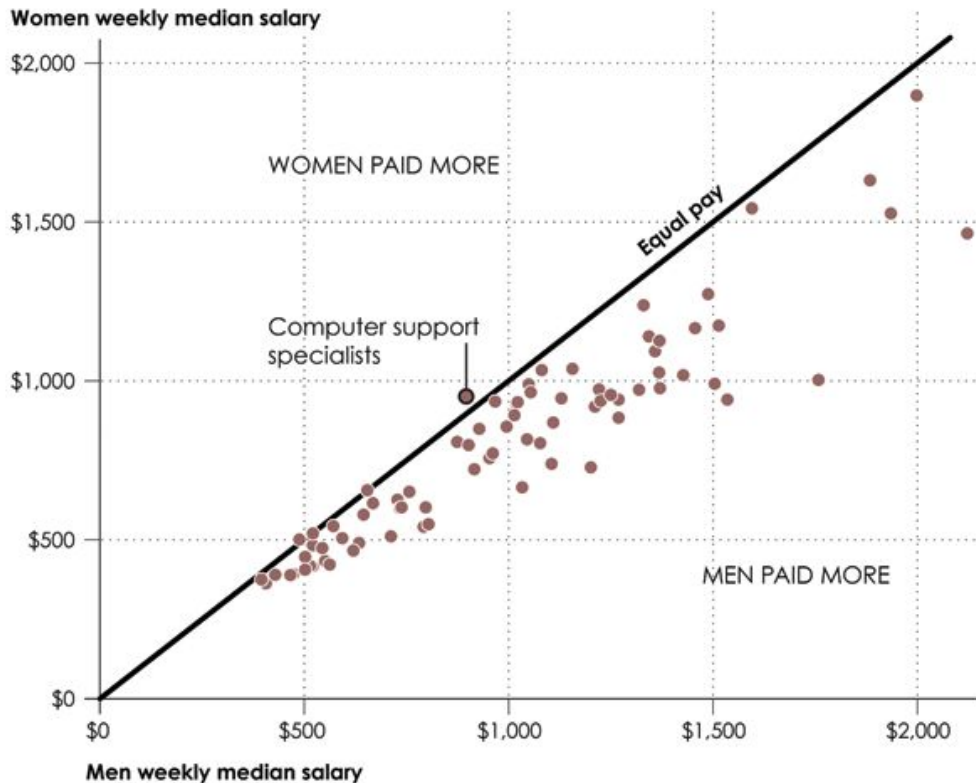


Source: Bureau of Labor Statistics

Contexto

Un gráfico sin
contexto **¡no
tiene sentido!**

Gender pay gap in 2011

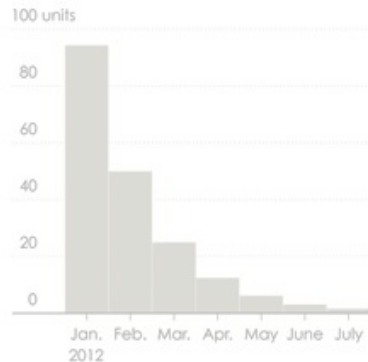


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Los ingredientes adecuados

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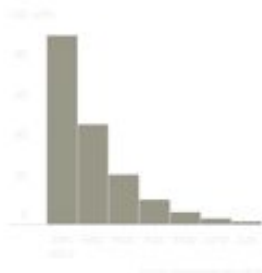
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Source: Somewhere reputable

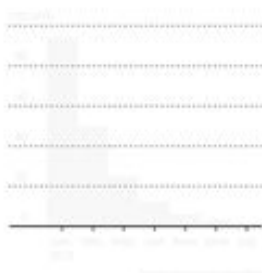
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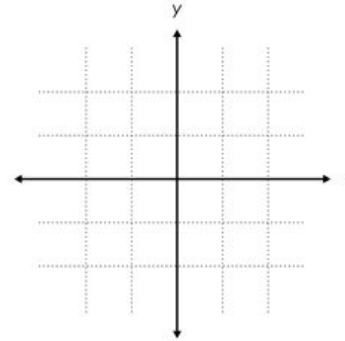
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De que estoy hablando

Sistema de coordenadas

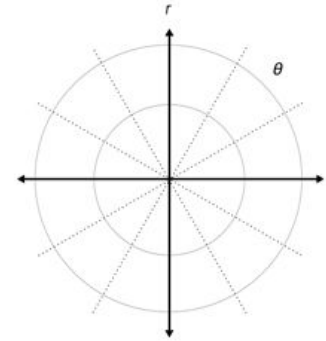
Cartesian

If you've ever made a graph, the x- and y-coordinate system will look familiar to you.



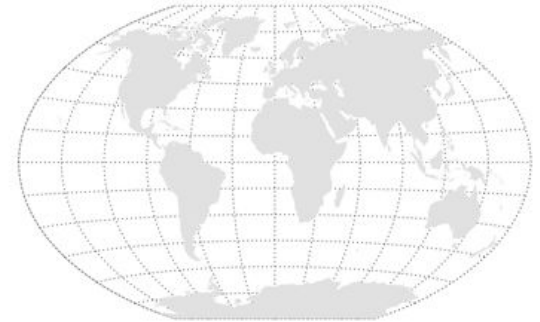
Polar

Pie charts use this system. Coordinates are placed based on radius r and angle θ .



Geographic

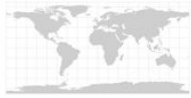
Latitude and longitude are used to identify locations in the world. Because the planet is round, there are multiple projections to display geographic data in two dimensions. This one is the Winkel tripel.



Sistema de coordenadas

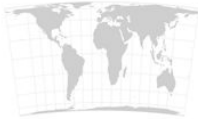
Equiangular

Typically used for thematic mapping, but doesn't preserve area or angle



Albers

Scale and shape not preserved; angle distortion is minimal



Mercator

Preserves angles and shapes in small areas, making it good for directions



Lambert conformal conic

Better for showing smaller areas and often used for aeronautical maps.



Sinusoidal

Preserves area; useful for areas near the prime meridian



Polyconic

Was often used to show US in the mid-1900s; little distortions in small areas near meridian



Winkel Tripel

Minimized area, angle, and distance distortion; good choice for world map



Robinson

A compromise between preserving areas and angles; good to show world map



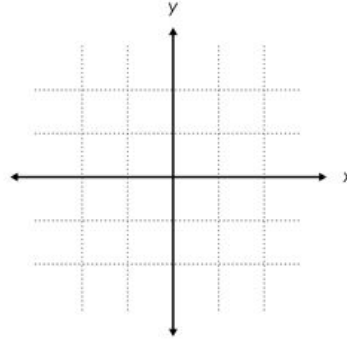
Orthographic

Representing a 3-D object in 2-D, need to rotate to area of interest



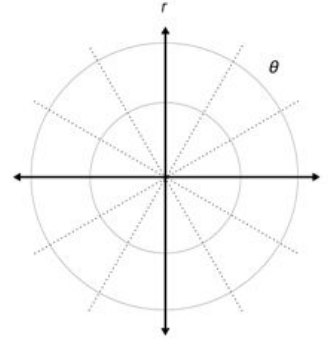
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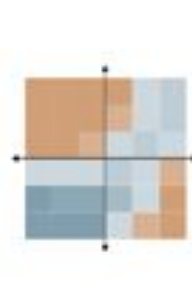
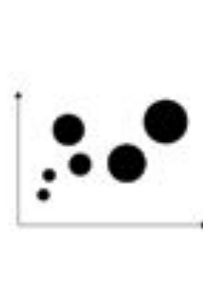
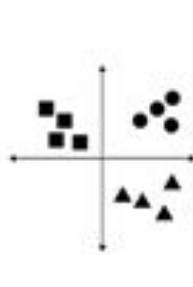




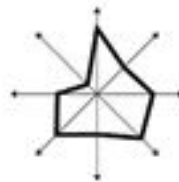
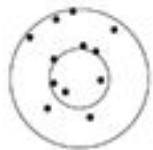




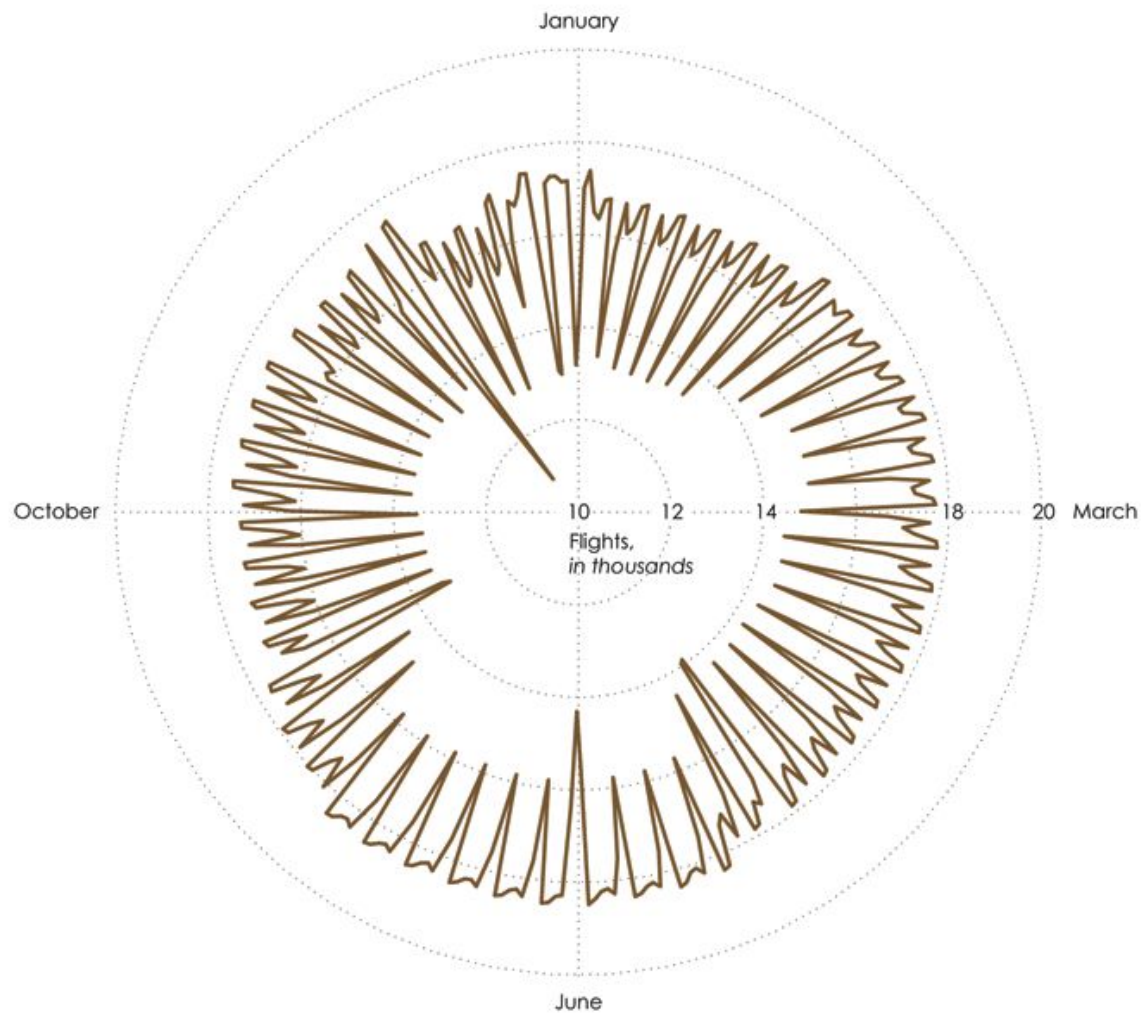
Coordenadas cartesianas



Coordenadas polares



Coordenadas polares



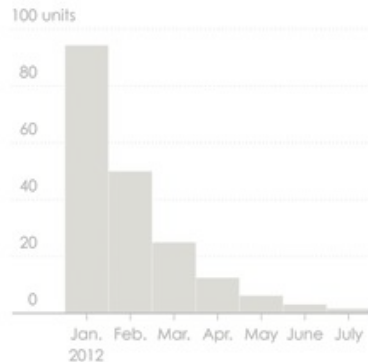
Coordenadas geográficas



Los ingredientes adecuados

Title of this Graph

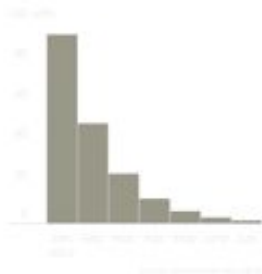
A description of the data or something worth highlighting to set the stage.



Source: Somewhere reputable

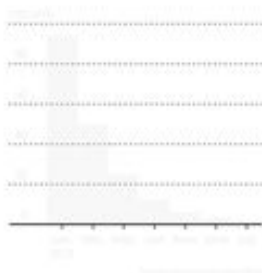
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Como mapear números visualmente

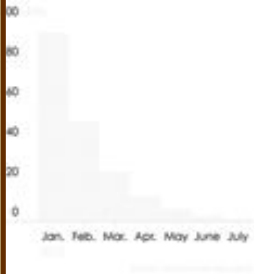
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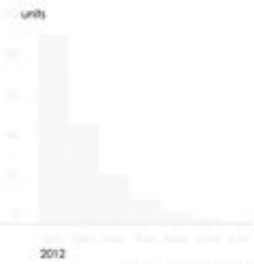
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Context

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De que estoy hablando

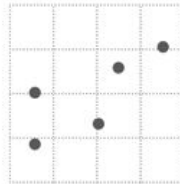
Señales visuales

Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

Position

Where in space the data is



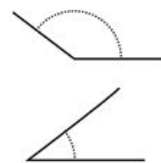
Length

How long the shapes are



Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



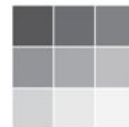
Volume

How much 3-D space



Color saturation

Intensity of a color hue

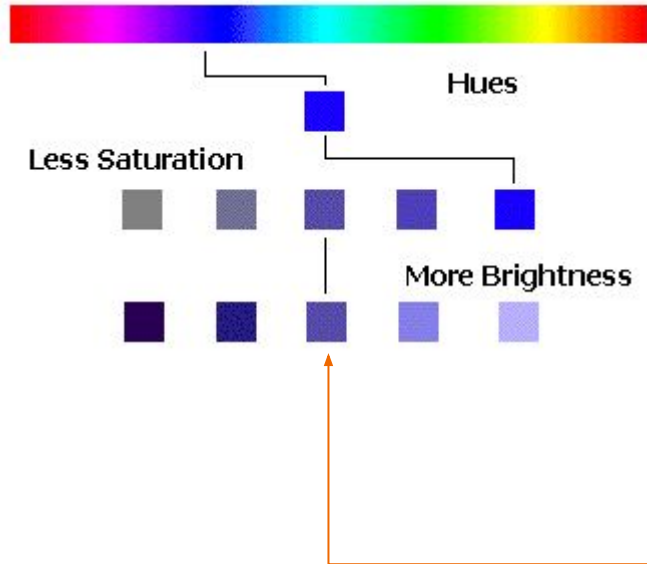


Color hue

Usually referred to as color



Señales visuales

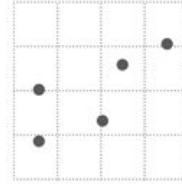


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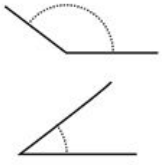
Length

How long the shapes are



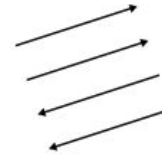
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space

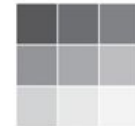


Volume

How much 3-D space



Color saturation or brightness



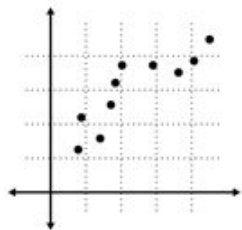
Color hue

Usually referred to as color

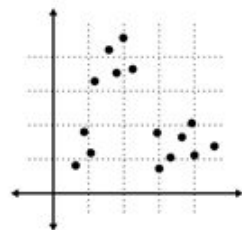


Señales visuales

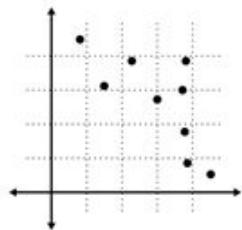
Upward trend



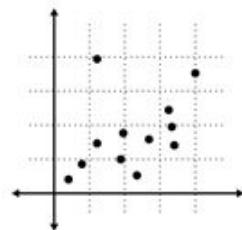
Clustering



Downward trend



Outlier

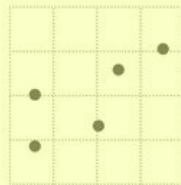


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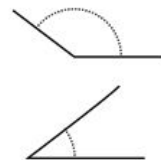
Length

How long the shapes are



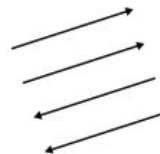
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



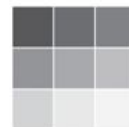
Volume

How much 3-D space



Color saturation

Intensity of a color hue



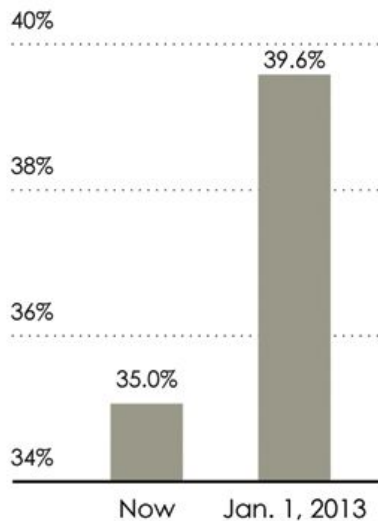
Color hue

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Señales visuales

Axis starting at 34 percent

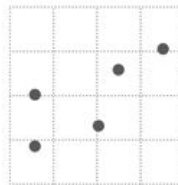


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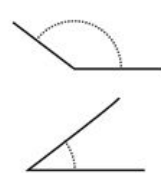
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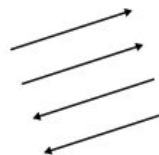
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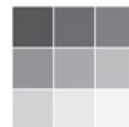
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Color hue

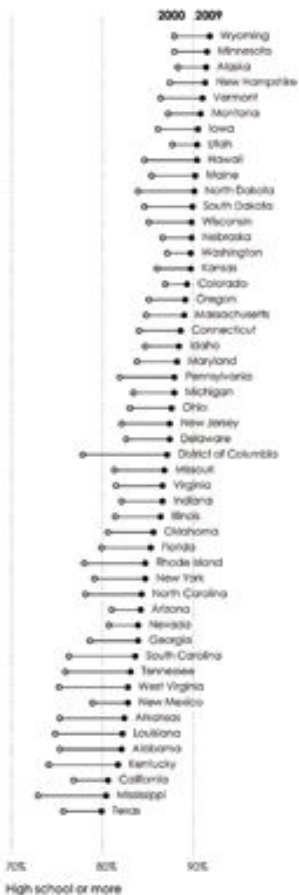
Usually referred to as color



Señales visuales

Greatest to least, endpoint

Focus on most recent numbers

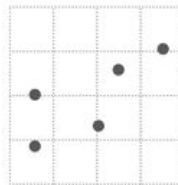


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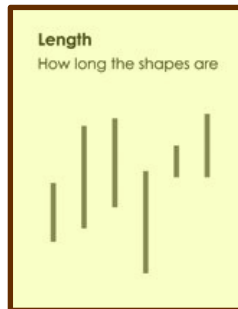
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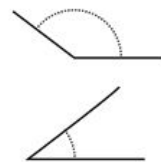
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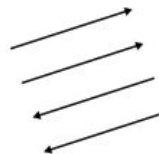
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Area

How much 2-D space



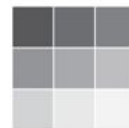
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Color saturation

Intensity of a color hue

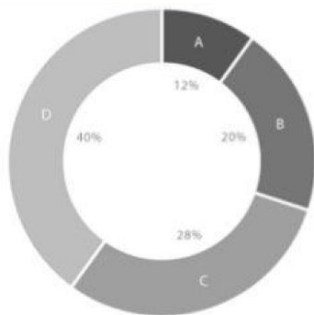


Color hue

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Señales visuales



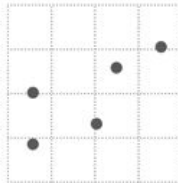
Donut charts!

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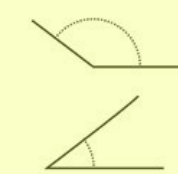
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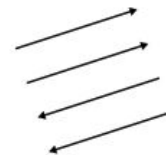
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Rotation between vectors



Direction

Slope of a vector in space



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Area

How much 2-D space



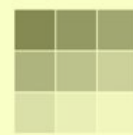
Volume

How much 3-D space



Color saturation

Intensity of a color hue



Color hue

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Señales visuales

Direction in a time series

20 metric tons of CO2 per capita in Australia



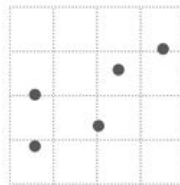
Source: The World Bank

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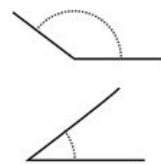
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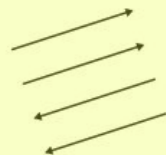
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

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Area

How much 2-D space



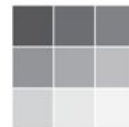
Volume

How much 3-D space



Color saturation

Intensity of a color hue

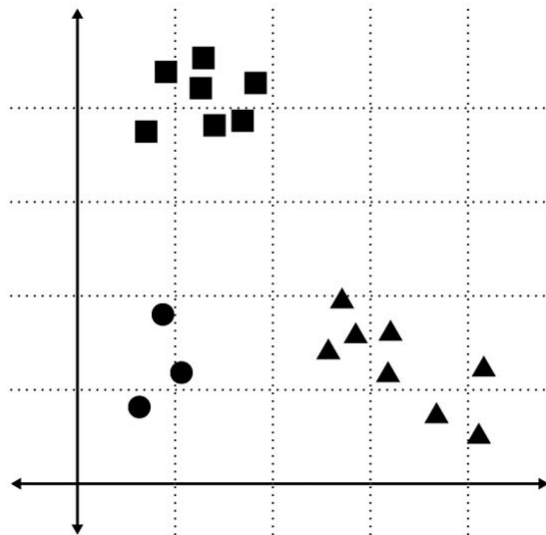


Color hue

Usually referred to as color



Señales visuales

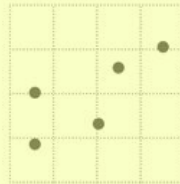


Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

Position

Where in space the data is



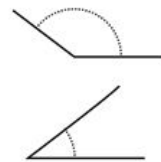
Length

How long the shapes are



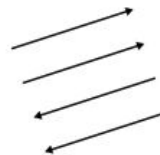
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



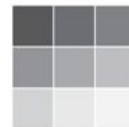
Volume

How much 3-D space



Color saturation

Intensity of a color hue

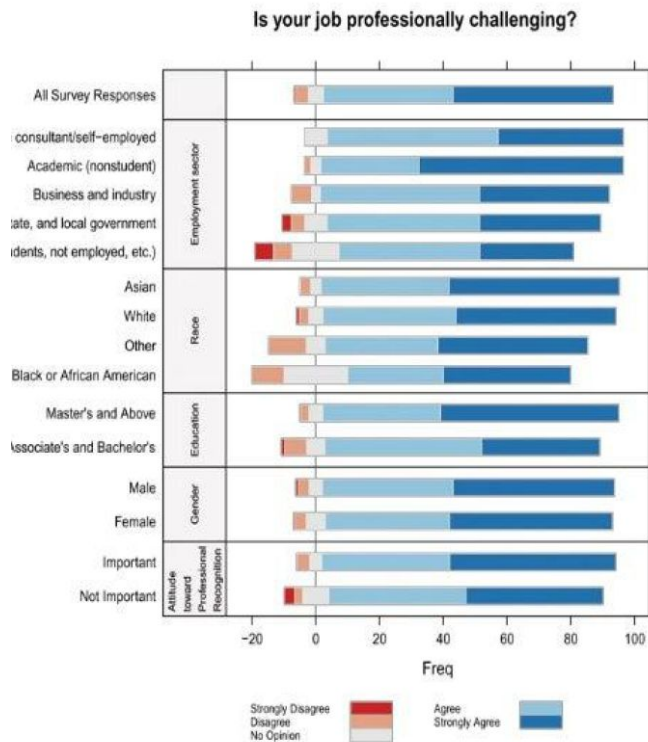


Color hue

Usually referred to as color



Señales visuales

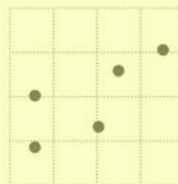


Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

Position

Where in space the data is



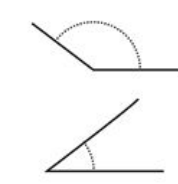
Length

How long the shapes are



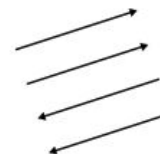
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



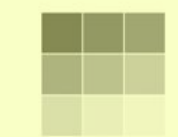
Volume

How much 3-D space



Color saturation

Intensity of a color hue



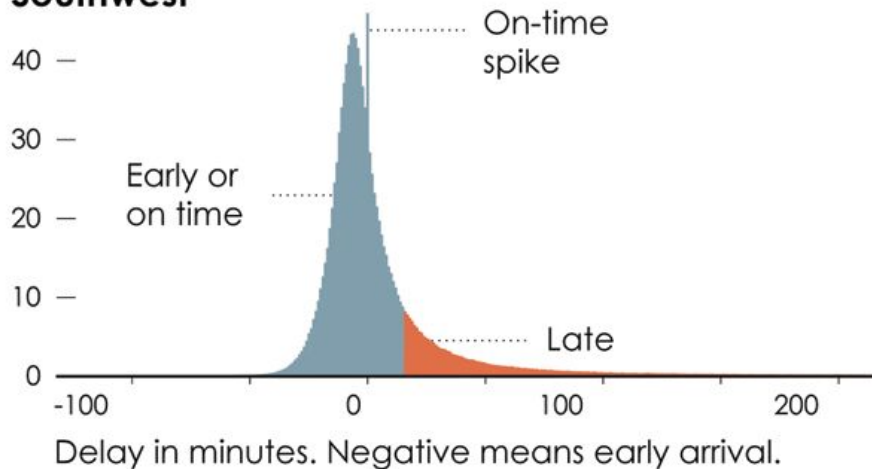
Color hue

Usually referred to as color



Señales visuales

Southwest

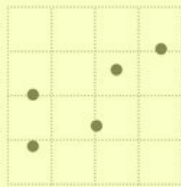


Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

Position

Where in space the data is



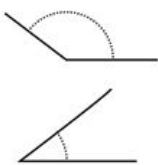
Length

How long the shapes are



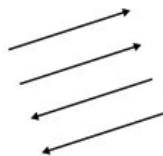
Angle

Rotation between vectors



Direction

Slope of a vector in space



Shapes

Symbols as categories



Area

How much 2-D space



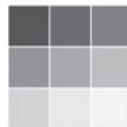
Volume

How much 3-D space



Color saturation

Intensity of a color hue

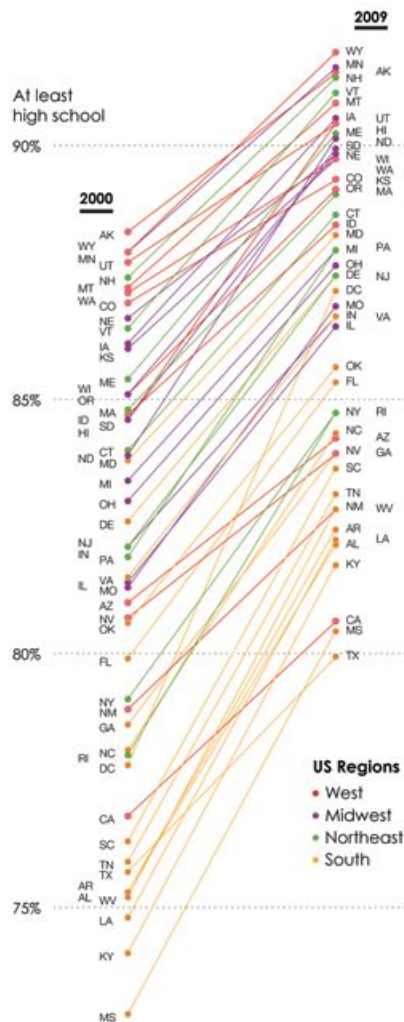


Color hue

Usually referred to as color



Señales visuales

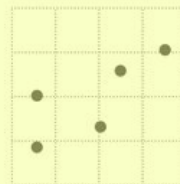


Visual cues

When you visualize data, you encode values to shapes, sizes, and colors.

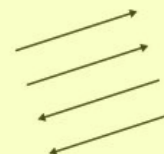
Position

Where in space the data is



Direction

Slope of a vector in space



Volume

How much 3-D space



Length

How long the shapes are



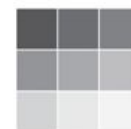
Shapes

Symbols as categories



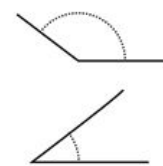
Color saturation

Intensity of a color hue



Angle

Rotation between vectors



Area

How much 2-D space

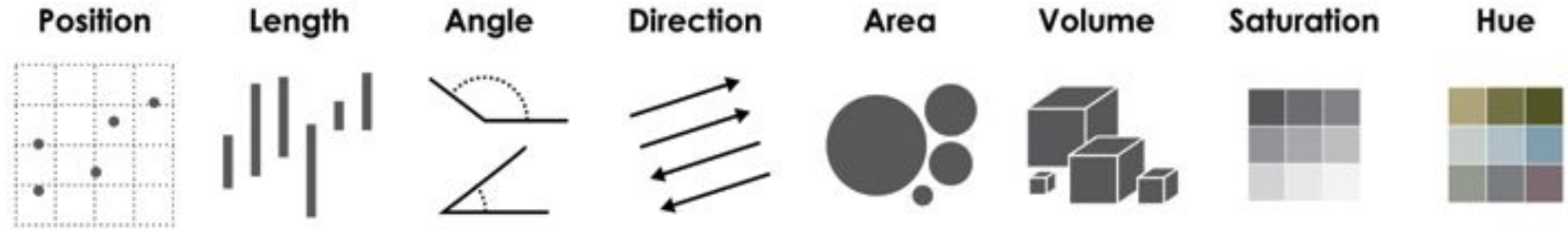


Color hue

Usually referred to as color

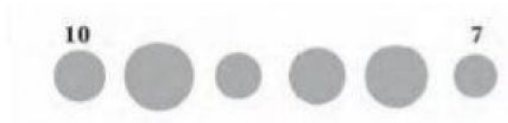


Señales visuales



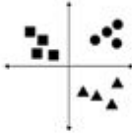
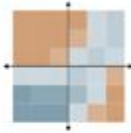
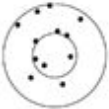
◀ More accurate

Less accurate ▶

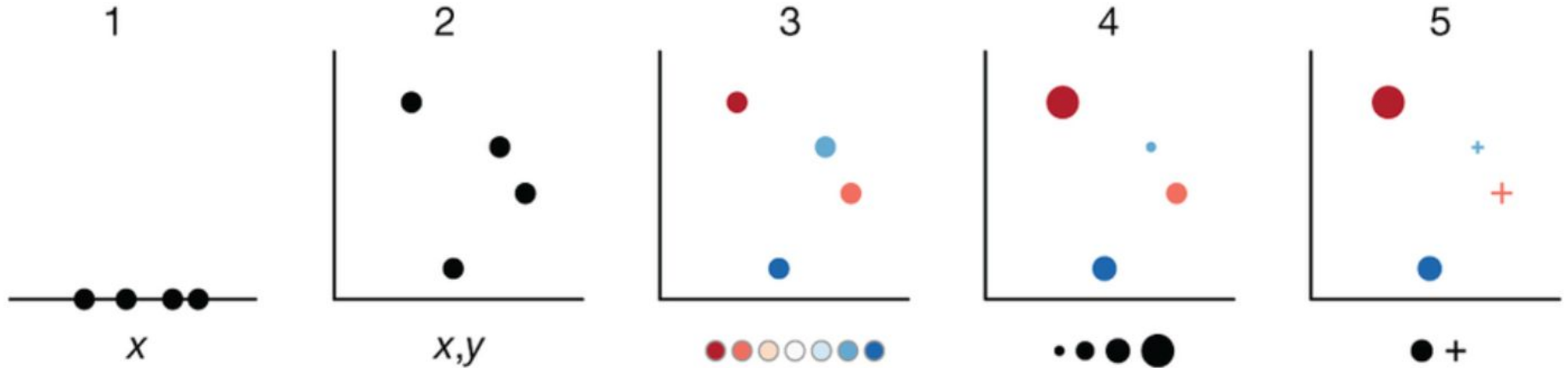


Combinación de ingredientes

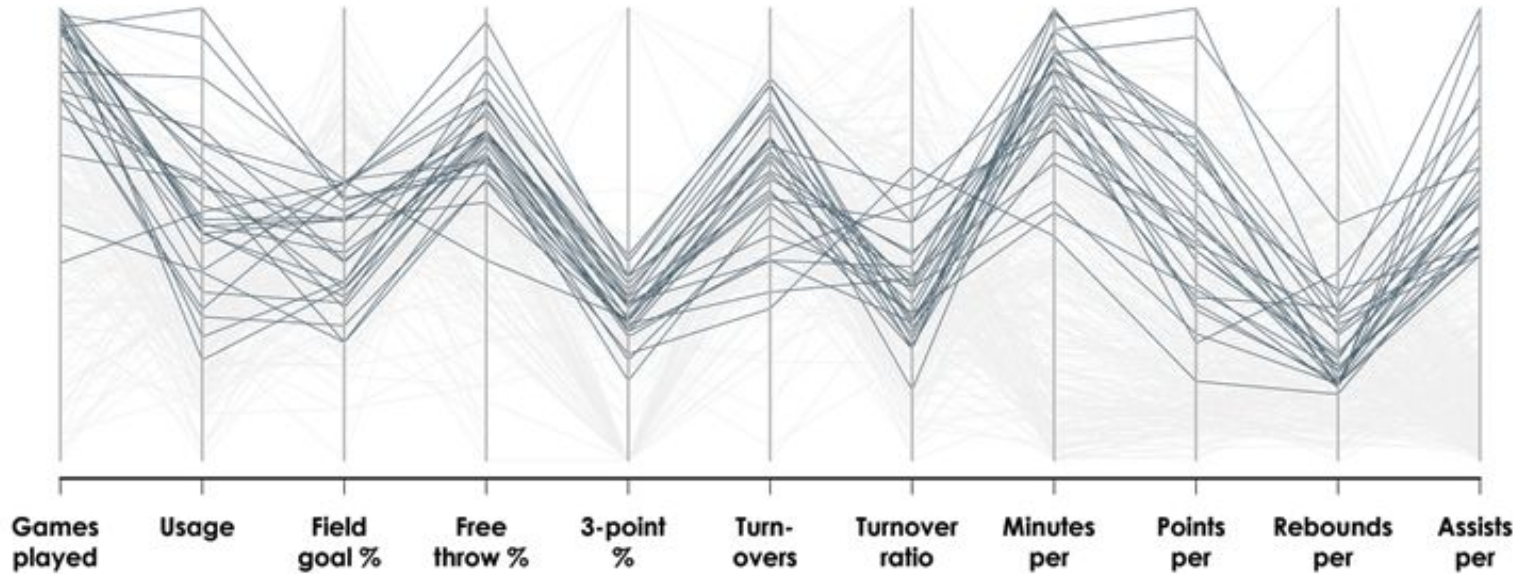
Visual cues

	Position	Length	Angle	Direction	Shapes	Area or Volume	Color
Coordinate systems							
Cartesian							
Polar							
Geographic							

Multidimensionalidad



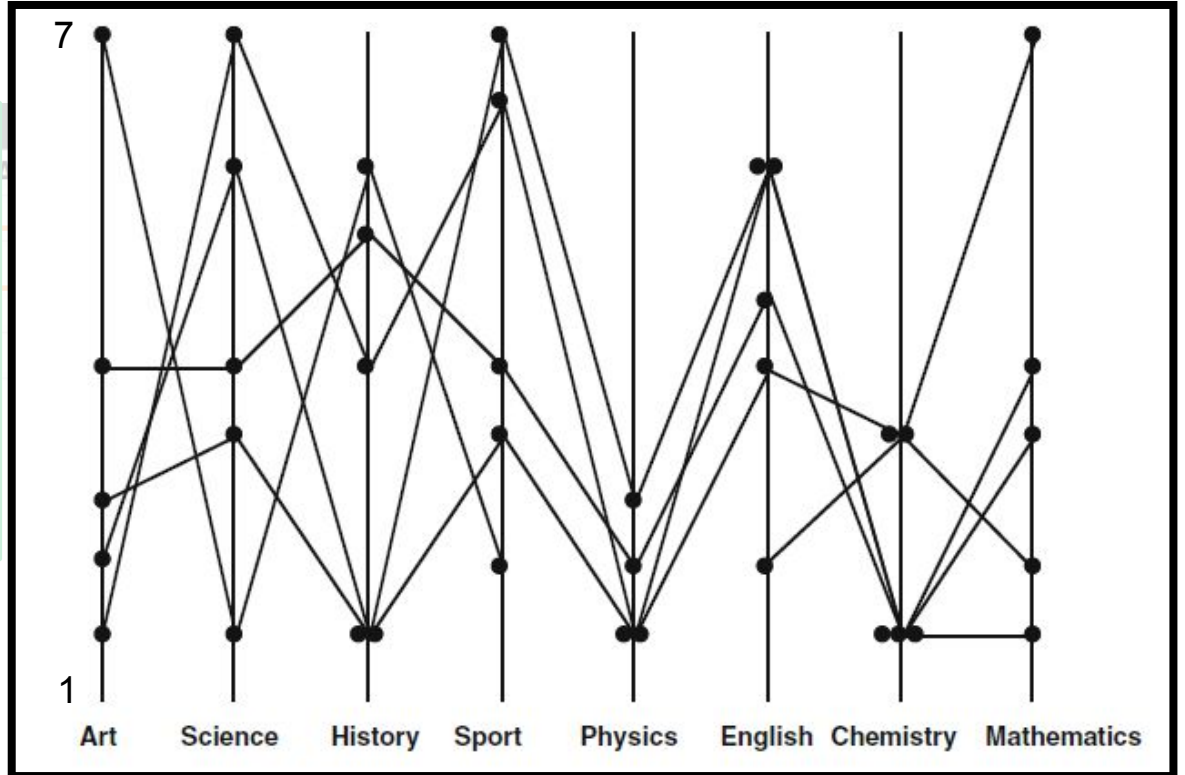
Multidimensionalidad



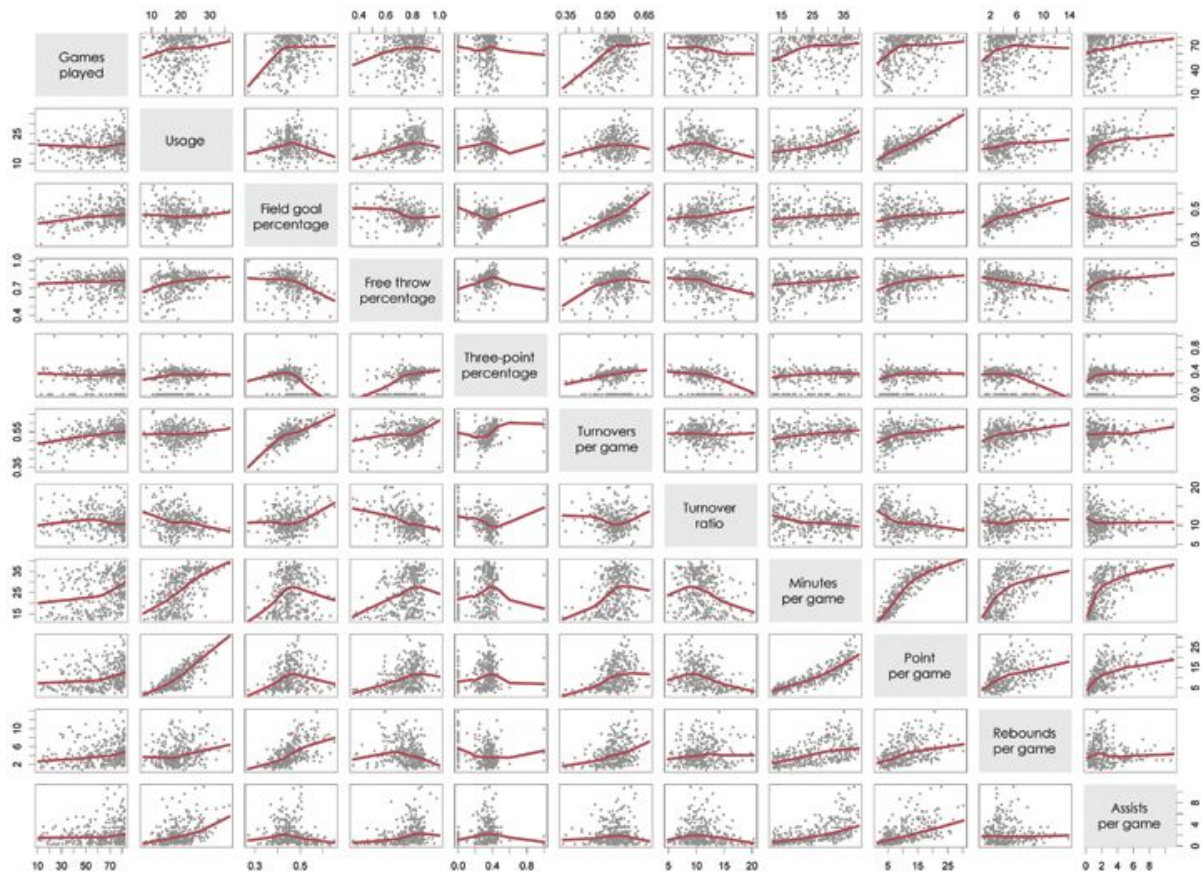
Multidimensionalidad

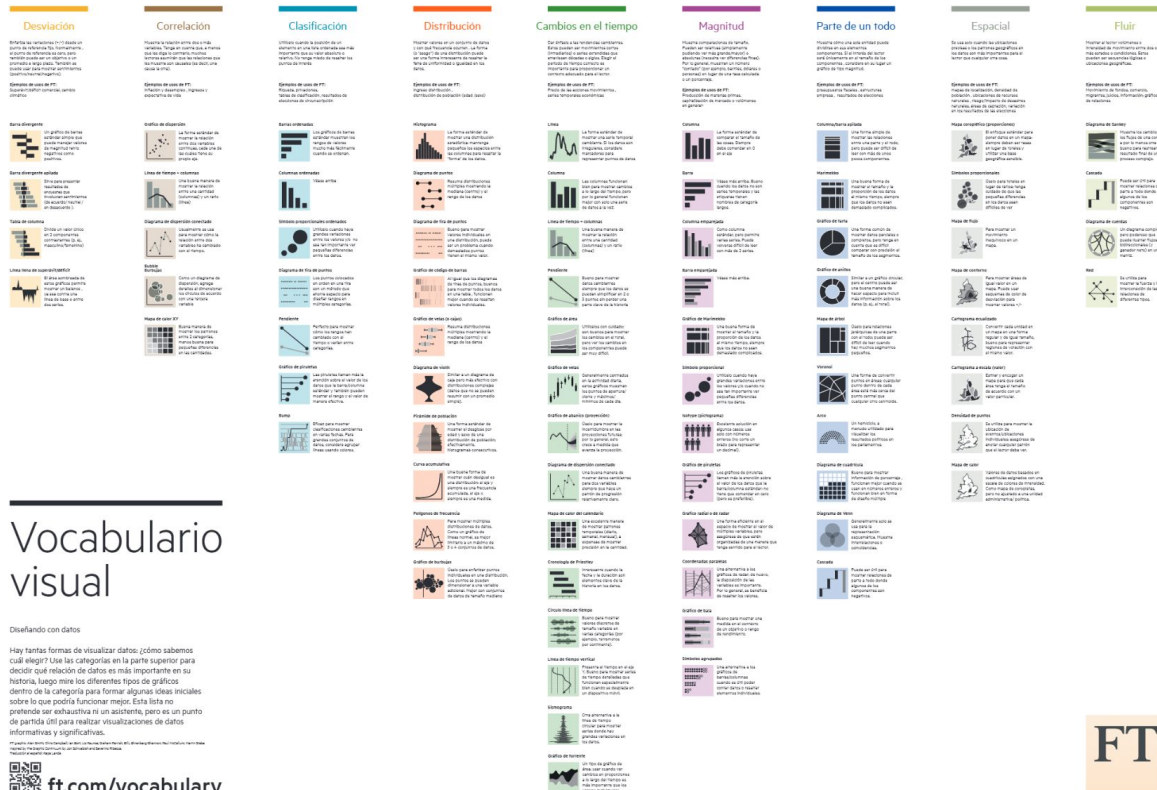
	A	B	C
1	Estudiante	Chimica	Fisica
2	Juan	5.7	4.6
3	Pedro	5.4	5.8
4	Vicente	5.5	5.4
5	Alicia	6.9	4.5
6	Silvia	2.2	5.3

Atributo



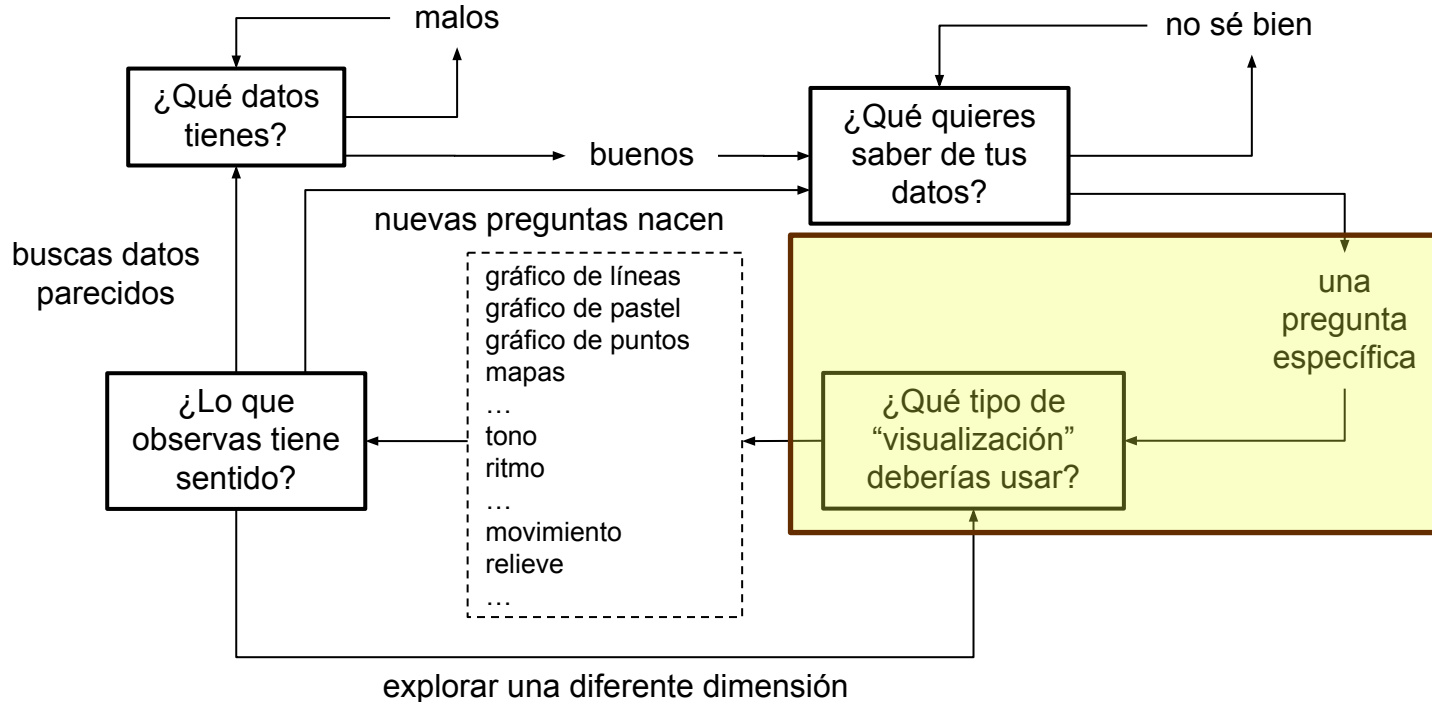
Multidimensionalidad





[chart-doctor/visual-vocabulary/Visual-vocabulary-es.pdf](#) at [main · Financial-Times/chart-doctor \(github.com\)](#)

Preguntas e ingredientes



Preguntas
posibles

Conceptos
estadísticos

Posibles
visualizaciones

¿Cuál _____ es el
mejor y el peor?

Máximos y
mínimos



¿Cómo ha
cambiado _____ a
lo largo del tiempo?

Patrones
temporales



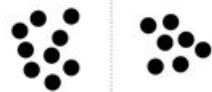
¿Qué _____ se
destaca del resto?

Valores
atípicos



¿Qué hace que
_____ sea diferente
de _____?

Agrupamiento



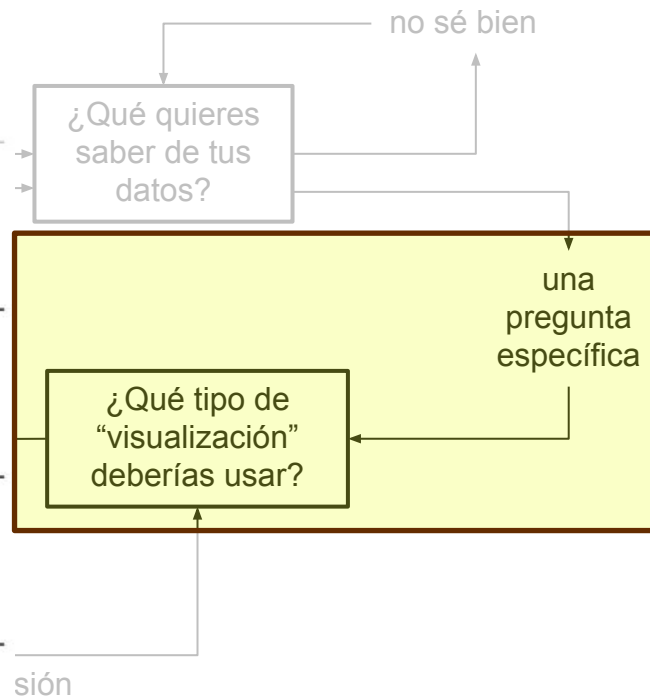
¿Cómo están
relacionados _____
y _____ entre sí?

Correlación



¿Cuál es la
descomposición de
_____?

Distribuciones



**¡Experimentar, probar, y
evaluar alternativas es clave!**

Interrogación 1

Interrogación 1 - Visualización de Información - Un ejemplo

IIC2026

Visualización de Información

Alessio Bellino